

**LETTER** **OPEN ACCESS**

# Epidemic Forecasting via Hybrid Deep Learning With Unified Visibility and Temporal Graphs Under Stochastic Noise Modelling

Arman Kavooosi Ghafi<sup>1</sup>  | Ali Pirkhedri<sup>2</sup> | Samira Akhbarifar<sup>3</sup> | Mohammad Hossein Shafiabadi<sup>4</sup>
<sup>1</sup>Department of Computer Engineering, Bo.C., Islamic Azad University, Borujerd, Iran | <sup>2</sup>Department of Computer Engineering, Mari.C., Islamic Azad University, Marivan, Iran | <sup>3</sup>Department of Computer Engineering, Pa.C., Islamic Azad University, Parand, Iran | <sup>4</sup>Department of Computer Engineering, Isl.C., Islamic Azad University, Islamshahr, Iran

**Correspondence:** Ali Pirkhedri ([alipirkhedri@gmail.com](mailto:alipirkhedri@gmail.com))

**Received:** 10 September 2025 | **Revised:** 6 January 2026 | **Accepted:** 16 March 2026

## ABSTRACT

Epidemic forecasts are unreliable when surveillance data are noisy or incomplete and when underreporting and rapidly changing population behaviour distort observed incidence, degrading the stability of conventional statistical and deep-learning models. We aim to develop an interpretable, uncertainty-aware forecasting pipeline that remains robust under data corruption and is practical for real-time use. We convert COVID-19 incidence into multilayer temporal graphs: global cumulative counts are differenced to daily incidence, normalised, and segmented into overlapping 30-day windows; for each window, we build a visibility graph from the empirical series and a matched-length visibility graph from stochastic simulations (fractional Brownian motion and Lévy-type dynamics) to represent reporting and behavioural randomness. We fuse the graphs (weighted edge averaging), extract compact descriptors (mean degree, clustering coefficient, entropy) and train a lightweight regressor to predict the 7-day-ahead average incidence. On the Johns Hopkins COVID-19 dataset, the method outperforms ARIMA, LSTM and standard GCN baselines (MAE = 0.0558; RMSE = 0.0709). Stress tests with noise and missingness and ablations show that stochastic augmentation and graph fusion materially improve robustness, while a cloud-oriented deployment reduces inference time by >60% and memory usage by 35%, enabling low-latency monitoring for timely public-health decision-making.

## 1 | Introduction

The global impact of epidemics — spanning both biological outbreaks and their digital counterparts — has underscored the critical importance of predictive modelling in mitigating public-health and cybersecurity crises [1, 2]. Human mobility and interconnected digital infrastructures accelerate virus dissemination, strengthening the need for forecasting methods that remain reliable under real-world complexity [3]. Classical epidemic modelling has traditionally relied on mechanistic compartmental frameworks, including susceptible–infected–recovered (SIR) and susceptible–infected–susceptible models [4]. While foundational,

these formulations often assume population homogeneity and simplified mixing patterns, limiting their fidelity when transmission is shaped by heterogeneous contacts, abrupt policy interventions, seasonal effects and reporting delays. As a result, the static structure and strong assumptions of compartmental models can restrict their applicability in operational settings characterised by dynamic and unpredictable interactions [5].

Motivated by these limitations, recent work increasingly applies machine learning—particularly recurrent neural networks (RNNs) and graph neural networks (GNNs) — to capture spatiotemporal dependencies beyond conventional time-series

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2026 The Author(s). *Healthcare Technology Letters* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology.

analyses [6–9]. Although such methods can improve predictive accuracy, especially when spatial interactions and auxiliary signals are available [9], reliability and interpretability remain challenging in practice. Surveillance data are frequently noisy, incomplete and biased due to underreporting, heterogeneous testing capacity, changing case definitions and irregular reporting schedules, which can create substantial deviations between predicted and observed trajectories [10, 11]. Moreover, epidemic spread is strongly influenced by stochastic human behaviour and mobility; without explicit uncertainty modelling, forecasting systems can overfit historical regimes and degrade under novel conditions, policy shifts, or emerging variants [12, 13].

A further limitation concerns how the contact structure is represented. Real-world networks evolve with changes in social behaviour, policy interventions and mobility patterns [14]. Many forecasting pipelines either assume a static network or rely on externally specified graphs (e.g., mobility matrices) that may be sparse, delayed, biased, or misaligned with the temporal resolution of case data. Although complex-network theory — particularly scale-free and small-world structures — offers insights into epidemic thresholds and superspreader dynamics [15], a recurring gap remains: time-series signals and network topology are often handled separately, which can constrain predictive granularity and obscure how structural shifts in the epidemic curve map to evolving transmission dynamics [16]. In addition, uncertainty is frequently handled implicitly (or not at all) and interpretability remains limited when learned representations cannot be connected to observable structural signatures in the epidemic trajectory.

**Problem statement.** Operational forecasting must predict near-term incidence from surveillance time series that are noisy, incomplete and nonstationary, while the effective contact structure can change rapidly. Existing methods can be fragile in this setting because they rely on static or externally provided graphs, fail to model stochastic variability explicitly and provide limited interpretability for decision support.

To address these gaps, this study proposes a unified, uncertainty-aware epidemic forecasting framework that integrates time-series structure, stochastic modelling and complex network representations within a single pipeline [17]. We convert epidemic time series into visibility graphs, where local extrema, bursts and regime shifts are expressed as topology rather than purely numerical fluctuations. In parallel, we generate synthetic stochastic sequences using fractional Brownian motion (fBm) and Lévy processes to model irregularity, long-range dependence and heavy-tailed perturbations associated with reporting and behavioural variability [3, 11]. Each synthetic sequence is also mapped to a visibility graph, yielding a noise-aware graph layer aligned with the empirical graph in length and temporal resolution. We then fuse the empirical and stochastic graph layers into a multilayer temporal representation that preserves salient deterministic structure while enriching it with uncertainty-aware connectivity. The resulting sequence of fused graphs forms a temporal graph representation of epidemic evolution and can be coupled with temporal graph learning (e.g., temporal graph networks [TGNs]) to model dynamic patterns across time windows. When available, biomedical imaging features can be incorpo-

rated as node- or window-level attributes, enabling forecasts that reflect both population-level transmission dynamics and clinically informative markers.

The proposed pipeline is designed for practical deployment. Scalable cloud computing environments provide elastic compute, distributed storage and real-time processing for high-volume epidemic data streams, yet many forecasting models remain difficult to operationalise under latency and resource constraints. Our modular design supports parallel execution of graph construction, stochastic simulation and feature extraction, enabling scalable training and low-latency inference in cloud-native settings. Cloud–edge compatibility further supports geographically distributed operation by allowing local preprocessing while executing graph fusion and temporal learning in the cloud.

This work builds on evidence that robust forecasting benefits from modelling noise, mobility and evolving structure. Prior studies investigated learning epidemic graphs under noisy conditions [18], classical frameworks established foundational infectious-disease dynamics [19] and network-based analyses emphasised topology in outbreak behaviour [4]. Modern approaches incorporated epidemiological metrics and interventions such as social distancing [20, 21] and mobility-aware modelling have proven vital for simulating spatial spread [22]. However, jointly integrating time-series structure, stochastic variability and adaptive network representations in a deployment-ready framework remains an open practical need, which this study aims to address.

Unlike BDSTGNN [23], which couples a backbone graph with temporal dynamics but does not explicitly construct uncertainty-aware topology from the epidemic signal itself, our framework converts each time window into a visibility graph and augments it with a generative stochastic counterpart (e.g., Brownian/fBm/Lévy-type noise), enabling window-level uncertainty injection directly at the graph-construction stage. In contrast to MepoGNN [24], which focuses on coupling GNNs with mechanistic SIR parameterisation, our approach remains model-agnostic to compartmental assumptions and instead learns from topological signatures of the observed epidemic trajectory enriched by stochastic perturbation. Compared with multi-scale STGNN designs such as MSGNN [25], our method emphasises data-driven graph formation and uncertainty-aware fusion that can operate with minimal dependence on externally curated spatial graphs. Quantitatively, on the Johns Hopkins COVID-19 dataset, our model achieves  $\text{RMSE} = 0.0709$ , improving upon long short-term memory (LSTM) and graph convolutional network (GCN) baselines by 10.4% and 20.0%, respectively and maintaining superior robustness under Gaussian noise ( $\text{RMSE} = 0.0779$ ; 18.7% lower than LSTM) and missingness ( $\text{RMSE} = 0.0763$ ; 22.5% lower than LSTM). Finally, the cloud-native deployment reduces inference latency from 14.5s to 4.7s (67.6%) and memory from 790 to 512 MB (35.2%), underscoring the method's practical scalability.

The main contributions of this work are as follows:

- We propose a novel hybrid deep learning framework that integrates temporal graph modelling with medical imaging features to improve the accuracy and robustness of epidemic forecasting.

- We develop a multilayer visibility-graph construction strategy that combines empirical epidemic time series with stochastic noise modelling (e.g., fBm and Lévy-type processes) to explicitly account for uncertainty and reporting variability.
- We introduce a graph fusion mechanism that produces an uncertainty-aware temporal graph representation, improving robustness to noisy and incomplete inputs.
- We validate the proposed framework on real-world COVID-19 time-series data and demonstrate consistent improvements over classical statistical baselines and state-of-the-art deep learning models.
- We present a cloud-native, modular deployment design that enables parallel graph construction and low-latency inference, supporting scalable real-time epidemic monitoring.

The remainder of this paper is organised as follows: Section 2 reviews related research; Section 3 presents the proposed methodology; Section 4 reports the experimental results; Section 5 discusses implications and limitations; and Section 6 concludes the paper.

## 2 | Related Work

Recent epidemic-forecasting research has shifted from purely mechanistic compartmental models toward hybrid deep learning and spatiotemporal graph learning, motivated by the need to represent mobility-driven coupling between regions, capture non-stationary temporal trends and remain robust to data noise and reporting artefacts. A central limitation of many early deep models was treating regions as independent time series (or using static adjacency), whereas real outbreaks evolve on time-varying, partially observed contact and mobility networks.

Mahmud et al. [1] proposed an early mobility-informed hybrid that combines graph learning with recurrent temporal modelling (GNN + RNN) to exploit inter-regional connectivity while preserving sequential dynamics. Although effective as a pragmatic fusion, such GNN-RNN pipelines typically inherit RNN drawbacks (error accumulation and sensitivity to noisy inputs) and do not explicitly formalise stochastic observation noise.

Zhu et al. [6] introduced a graph-attention-based spatiotemporal framework (GAST) that uses attention to weight influential neighbours and better capture heterogeneous spread patterns. Attention improves interpretability by revealing which nodes/edges dominate forecasts; however, attention-based STGNNs still commonly assume that noise is handled implicitly by the training loss, rather than through an explicit stochastic layer.

A major line of work couples deep spatiotemporal graph learning with mechanistic dynamics. Wang and Jin [26] integrated graph-transformer mechanisms with a metapopulation SIR formulation (M-Graphormer) to support parameter estimation and multi-region prediction under mobility-induced dynamic structures, including settings with limited data quality. Similarly, Cao et al. [24] proposed MepoGNN, which embeds graph learning into a metapopulation SIR framework to learn epidemiological param-

eters and latent propagation graphs end-to-end, improving interpretability relative to black-box deep models. These mechanistic-deep hybrids are valuable for structure and explainability, yet they often remain deterministic in the observation model and rarely unify alternative temporal graph constructions (e.g., visibility graphs) with mobility graphs in a single representation.

Beyond epidemiology-only settings, Ghafi et al. [27] modelled spread patterns on complex temporal networks for both human and computational epidemics, emphasising time-dependent interactions on evolving graphs. This direction strengthens temporal-network realism, but it does not directly address how to fuse time-series-to-graph transformations (e.g., visibility graphs) with spatiotemporal diffusion graphs for forecasting under measurement noise.

Miller et al. [28] advanced knowledge-enhanced deep learning by injecting domain knowledge (e.g., knowledge graphs) into pandemic prediction to improve robustness and interpretability when historical data are sparse. Mao et al. [23] proposed BDST-GNN, combining a learned backbone graph with temporal graph learning to build dynamic structures efficiently, highlighting that relying purely on static or purely dynamic graphs can be suboptimal. While these approaches improve structure learning and robustness, they still typically treat noisy surveillance signals as nuisances during learning, not as a first-class stochastic process.

Several referenced works [29–32] extend deep learning to pandemic-adjacent tasks (tourism recovery planning, overcrowding visual sensing, SIR-based country analyses and chest X-ray classification), illustrating the breadth of COVID-19 analytics; however, they are generally not designed to unify multiview temporal graph representations for multi-region epidemic forecasting.

Comparative and hybrid time-series studies provide important baselines and design insights. Dairi et al. [33] benchmarked ML/DL methods for COVID-19 transmission forecasting, reinforcing that deep and hybrid models often surpass classical statistical baselines. Zhan et al. [34] proposed deep visibility series, showing that visibility-graph-driven representations can strengthen forecasting in non-epidemic domains; this is directly suggestive for epidemic time series, yet it remains under-explored for spatiotemporal epidemic graphs. Yenurkar and Mal [35] and Li and Ma [36] further demonstrated that hybridisation (e.g., CNN/LSTM and transformer + GCN) can improve COVID-19 forecasts, but these models still usually lack explicit stochastic observation modelling and do not unify visibility-based temporal graphs with mobility graphs. Mathonsi and van Zyl [37] highlighted hybrid statistical-deep approaches for multivariate forecasting and uncertainty quantification (e.g., prediction intervals), yet without graph-structured diffusion modelling. Rahmadani and Lee [38] integrated hybrid deep learning with metapopulation modelling for South Korea, reinforcing the value of mobility-aware parameter estimation. Vargas et al. [39] showed that combining heterogeneous signals (technical indicators + text) can lift predictive performance in finance — methodologically relevant to epidemic nowcasting with multimodal covariates. Abbasimehr and Paki [40] used Bayesian optimisation to improve deep forecasting configurations, but optimisation

alone does not resolve structural gaps in graph representation or stochastic noise modelling.

Recent STGNN advances address scale, causality and mobility. Qiu et al. [25] proposed MSGNN to learn multi-scale spatiotemporal dependencies and preserve long-range connectivity that single-scale graphs may miss. Wang et al. [41] introduced Causal-GNN, explicitly incorporating causal/mechanistic structure into GNN learning to improve robustness and interpretability. Despite these advances, a persistent limitation is that most frameworks commit to a single notion of ‘temporal graph’ (e.g., temporal edges over node histories) and do not integrate visibility-graph temporal topology as an additional, complementary view.

Finally, to provide perspective on hybrid efficiency, Radhi et al. [42] proposed a hybrid variational quantum circuit (VQC)–EfficientNet model for BloodMNIST classification, arguing that hybrid quantum–classical design can reduce complexity without sacrificing accuracy. While not an epidemic-forecasting study, it strengthens the broader methodological motivation for hybrid architectures that trade off expressiveness, efficiency and robustness — a motivation aligned with epidemic settings where models must be deployable and stable under noisy, non-stationary signals.

To provide a comprehensive overview of the current state of epidemic forecasting research, Table 1 summarises key recent studies that employ deep learning and graph-based approaches.

Table 1 details each study’s methodology, datasets, novel contributions, quantitative results, strengths and limitations. Research gaps clarified (and motivating this work): (1) Unified graph representation is missing — existing epidemic STGNNs typically model mobility/adjacency dynamics but do not fuse visibility-graph temporal topology with spatiotemporal diffusion graphs in a single framework [25, 33, 41, 42]. (2) Stochastic noise is under-modelled — most methods absorb reporting noise implicitly through training loss, rather than explicitly modelling observation/transition noise to improve robustness and calibrated uncertainty [6, 23, 37, 40]. (3) Multiview temporal structure is underexploited — attention, multi-scale learning and causal constraints help, yet they usually operate on one temporal graph construction and do not leverage visibility graphs for early regime-change detection in epidemic curves [25, 34, 41]. (4) Practical deployability vs. fidelity remains a tension — large-scale mobility STGNNs can be powerful but may require heavy preprocessing and high-quality mobility streams, motivating compact hybrid designs and noise-aware learning [1, 26].

These gaps directly motivate epidemic forecasting via hybrid deep learning with unified visibility and temporal graphs, where visibility-based temporal graphs complement mobility-driven spatiotemporal graphs and stochastic modelling improves stability, generalisation and uncertainty calibration under real-world reporting noise.

### 3 | Methodology

In response to the limitations of traditional epidemic forecasting methods — which often rely on static assumptions, ignore

stochastic variability and fail to integrate clinical indicators — we propose a novel hybrid deep learning framework that unifies temporal graph modelling with medical imaging analysis. Unlike conventional time-series or purely biomedical models, our approach captures both the dynamic transmission patterns of epidemics and the biological manifestations of infection. By leveraging multilayer visibility graphs constructed from real epidemic time series and stochastic noise processes and integrating these with CNN-extracted features from medical imaging data, the model simultaneously addresses the randomness of human behaviour and the clinical progression of disease. To support real-world scalability and operational resilience, the architecture adopts a deployment-agnostic design. The modular structure, comprising independent graph generation, image encoding and deep learning modules, enables efficient execution on distributed infrastructures. This design facilitates parallel processing, dynamic resource allocation and coordination across heterogeneous compute tiers, making the framework highly suitable for large-scale epidemic forecasting in latency-sensitive public health systems. This dual-source distributed architecture enhances robustness to noisy or incomplete data and enables adaptive, clinically relevant forecasting pipelines deployable in real-time outbreak scenarios.

The overall architecture of the proposed hybrid forecasting framework is illustrated in Figure 1. The process begins with the collection and preprocessing of epidemic time series and medical imaging data. Visibility graphs are constructed over sliding temporal windows and augmented with stochastic noise simulations to increase resilience against uncertainty. Feature vectors extracted from medical images are embedded into graph nodes and a deep learning model is trained on the fused temporal graph structures. Finally, the model is evaluated using standard forecasting metrics under both clean and noisy input conditions, in both local and cloud-executed settings.

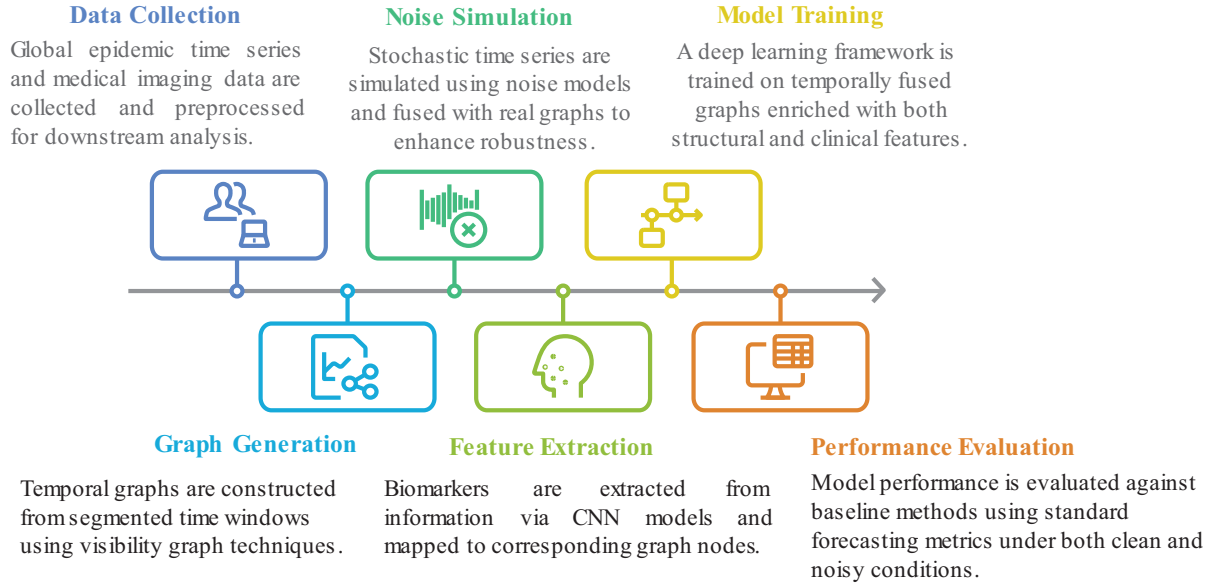
#### 3.1 | Overall Workflow and Novelty

Given an epidemic time series (and optional medical images), the proposed framework operates as follows (see Algorithm 1). (1) The cumulative case series is converted to daily incidence via first-order differencing and the resulting sequence is normalised (e.g., min–max scaling). (2) The normalised series is segmented into overlapping temporal windows of length  $W$  (e.g., 30 days) with stride  $s$ . (3) For each window, an empirical visibility graph  $G_t^{\text{emp}}$  is constructed by mapping each time point to a node and connecting pairs of nodes according to the visibility criterion.

(4) In parallel, a stochastic series of equal length is generated using fBm and/or Lévy-type dynamics and it is transformed into a noise visibility graph  $G_t^{\text{noise}}$ . (5) The two graphs are fused to form a multilayer uncertainty-aware graph  $G_t$  using a selected fusion rule (e.g., weighted edge averaging with threshold  $\tau$ ). (6) The fused graph sequence  $\{G_t\}$  is then used to produce predictions: either (a) graph structural descriptors (e.g., average degree, clustering coefficient and graph entropy) are extracted to form a feature vector, or (b) the fused temporal graphs are directly provided to the temporal graph learner. (7) The regression target for each window is defined as the  $H$ -step-ahead outcome (e.g., the 7-day forward average incidence). Model training uses a temporal

**TABLE 1** | Overview of key studies on deep learning and graph-based methods for epidemic forecasting, summarising approaches, datasets, innovations, results, advantages and limitations.

| Reference                 | Method                                                 | Dataset                                          | Novelty                                                       | Quantitative results                               | Advantages                                   | Limitations                              |
|---------------------------|--------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------|----------------------------------------------|------------------------------------------|
| Wang and Jin [26]         | M-graphormer: graph transformer + metapopulation SIR   | Real outbreak data (multi-region)                | First to combine GNN transformers with epidemiological models | Higher accuracy under noisy data (multi-region)    | Handles low-data regions, dynamic adaptation | Complex model structure                  |
| Cao et al. [24]           | MepoGNN: GNN + mechanistic SIR                         | COVID-19 multi-region datasets                   | Learning epidemiological parameters via GNNs                  | Better predictive performance and interpretability | High interpretability + accuracy             | Requires large graph-structured datasets |
| Miller et al. [28]        | Knowledge-enhanced deep learning with knowledge graphs | Synthetic + historical pandemic data             | Injects domain knowledge for interpretability                 | Improved accuracy and robustness                   | Boosts robustness and interpretability       | Needs accurate prior knowledge           |
| Mao et al. [23]           | BDSTGNN: Static backbone + dynamic temporal graphs     | Influenza and COVID-19 data                      | Fusion of static and dynamic graphs                           | Reduced computation with high forecast accuracy    | Balances accuracy and efficiency             | Assumes availability of backbone graphs  |
| Dairi et al. [33]         | Comparison of ML methods (LSTM, hybrids, classical)    | COVID-19 case datasets (various regions)         | Systematic ML performance comparison                          | LSTM and hybrids outperform classical ML           | Empirical strength of hybrid models          | Overfitting risk on small datasets       |
| Zhan et al. [34]          | Deep Visibility Series for time series                 | Construction indices (transferable to epidemics) | Applies visibility graphs to forecasting                      | Promising for time series; no epidemic results     | Novel method; cross-domain potential         | Not validated on epidemic data           |
| Yenurkar and Mal [35]     | Hybrid CNN-LSTM deep learning model                    | COVID-19 India datasets                          | Combines CNN and LSTM for noisy data                          | Higher robustness to noise; better MAE/RMSE        | Better performance in noisy environments     | Needs larger validation datasets         |
| Li and Ma [36]            | Hybrid transformer and GCN model                       | COVID-19 global datasets                         | Improved transformer-GCN hybrid                               | Lower forecast errors vs LSTM/ARIMA                | Combines local and global dependencies       | Increased model complexity               |
| Mathonsi and van Zyl [37] | MES-LSTM: statistics + LSTM hybrid                     | COVID-19 multivariate datasets                   | Hybrid statistics and deep learning approach                  | Better multivariate prediction performance         | Captures uncertainty better than standard DL | Hybrid models harder to tune             |
| Rahmadani and Lee [38]    | Hybrid deep learning with human mobility data          | COVID-19 data South Korea                        | Human mobility in hybrid DL epidemic model                    | Outperformed standard SEIR models                  | Better response to mobility-based changes    | Mobility data availability constraints   |
| Vargas et al. [39]        | Deep learning with technical indicators + news data    | Stock market datasets (transferable)             | Technical and textual feature fusion                          | Improved predictive accuracy for financial data    | Integrates multiple information types        | Cross-domain generalisation uncertain    |
| Abbasimehr and Paki [40]  | Deep learning + Bayesian optimisation                  | COVID-19 case datasets                           | Bayesian optimisation with DL tuning                          | Better generalisation and tuning performance       | Optimises hyperparameters systematically     | Hyperparameter search is time-consuming  |



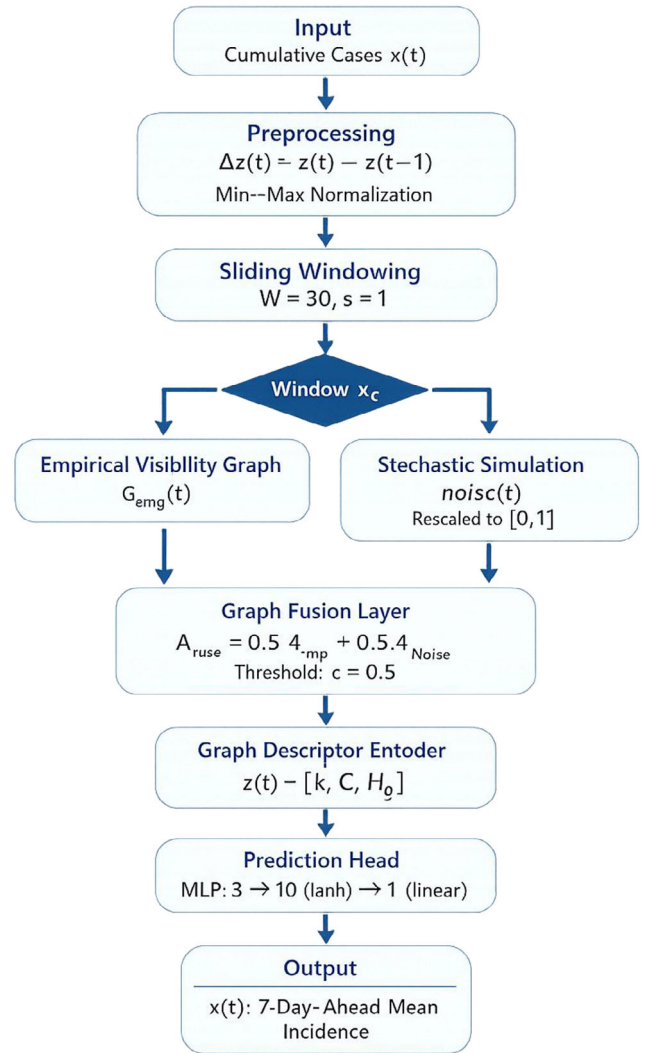
**FIGURE 1** | Overview of the proposed hybrid framework for epidemic forecasting.

**ALGORITHM 1** | Noise-Aware Visibility-Graph Fusion for Epidemic Forecasting.

- Input:** incidence series  $x$ , window length  $W$ , stride  $s$ , horizon  $H$ , fusion threshold  $\tau$   
**Output:** forecasts  $\hat{y}_t$
1. Compute daily incidence by differencing; normalise the series.
  2. Segment the series into windows  $\{x_t\}$  of length  $W$  with stride  $s$ .
  3. **For** each window  $x_t$ :
  4. Construct empirical visibility graph  $G_t^{\text{emp}}$ .
  5. Generate stochastic series  $n_t$  (fBm/Lévy) and construct  $G_t^{\text{noise}}$ .
  6. Fuse graphs  $G_t = f_{\text{fuse}}(G_t^{\text{emp}}, G_t^{\text{noise}}; \tau)$ .
  7. Form representation  $z_t$  (graph descriptors or direct graph input to Model).
  8. Define target  $y_t$  as the  $H$ -step-ahead (e.g., 7-day average) outcome
  9. **End For**
  10. Train the predictor on  $\{(z_t, y_t)\}$  using a temporal split; evaluate on held-out future windows.
  11. Return forecasts  $\hat{y}_t$  and compute evaluation metrics.

holdout split to prevent information leakage and performance is reported using standard forecasting metrics under clean and perturbed (noisy/incomplete) inputs.

The proposed architecture transforms epidemic time series into stochastic-enriched temporal graph representations via dual visibility-graph construction and graph fusion (see Figure 2). Compact structural descriptors extracted from each fused graph are then mapped to 7-day-ahead incidence forecasts using a



**FIGURE 2** | Architecture of the proposed hybrid TGN-VG forecasting framework.

lightweight neural prediction head. This hybrid design integrates interpretable graph-based encoding with efficient deep learning for robust epidemic forecasting.

### 3.2 | Preprocessing Step

Let  $\{\chi_t\}_{t=1}^T$  denote the raw epidemiological time series of cumulative confirmed case counts. We apply the following preprocessing steps to obtain a numerically stable incidence signal suitable for window-wise graph construction.

**Step 1:** We first compute the daily incidence series by first-order differencing:

$$\Delta x_t = x_t - x_{t-1}, t = 2, \dots, T \quad (1)$$

If negative values arise due to retrospective reporting corrections, we set  $\Delta x_t = \max(\Delta x_t, 0)$  to avoid non-physical incidence.

**Step 2:** The incidence series is scaled to  $[0,1]$  using min-max normalisation:

$$x'_t = \frac{\Delta x_t - \min(\Delta x)}{\max(\Delta x) - \min(\Delta x)} \quad (2)$$

This scaling improves numerical stability and ensures comparable ranges across time periods.

**Step 3:** To reduce high-frequency reporting artefacts (e.g., weekend effects), we optionally apply a simple moving average with a window length  $m$  (set to  $m = 7$  when smoothing is enabled):

$$\tilde{x}_t = \frac{1}{m} \sum_{i=0}^{m-1} x'_{t-i} \quad (3)$$

Unless stated otherwise, results are reported on the unsmoothed normalised series.

**Step 4:** Finally, the pre-processed sequence is segmented into overlapping windows of length  $W$  with stride  $s$ :

$$\mathbf{x}_t = \{x'_t, x'_{t+1}, \dots, x'_{t+W-1}\} \quad (4)$$

where  $W = 30$  days and  $s = 1$  day in our experiments. Each window is then used to construct the corresponding (empirical and stochastic) visibility graphs in the subsequent stage.

### 3.3 | Visibility Graph Construction

To capture the evolving structural dynamics of epidemic time series, the normalised data are partitioned into fixed-length sliding windows of length  $W = 30$  days (stride  $s = 1$  day) in our

experiments. This choice is supported by the hyperparameter sensitivity analysis reported in Table 2, where  $W \in \{14, 21, 30, 45, 60\}$  was evaluated under 5-fold cross-validation on the training split and  $W = 30$  provided the best validation trade-off. We avoid very short windows (e.g., 4–7 days) because they contain too few observations to induce stable and informative visibility-graph topology and are highly sensitive to weekly reporting periodicity and day-to-day reporting artefacts. Within each window, a visibility graph is constructed by mapping the time-series values to a graph structure in which each data point becomes a node and edges are formed based on geometric visibility criteria. This transformation encodes temporal patterns, local extrema and nonlinear dependencies inherent in epidemic progression. Consequently, each window yields a distinct visibility graph reflecting the underlying epidemiological behaviour during that period and the sequence of windows produces a temporally ordered graph stream representing the epidemic's evolution.

To encode temporal structures and local patterns present in epidemic progression, each pre-processed time-series segment is transformed into a visibility graph. In this graph representation, each data point within a temporal window is treated as a node and edges are established based on a geometric visibility criterion that reflects the relative magnitude and position of the points in the series. Specifically, two data points  $x_i$  and  $x_j$  (with  $i < j$ ) are said to have a visibility connection if and only if, for every intermediate point  $x_k$  (where  $i < k < j$ ), the following condition holds:

$$x_k < x_i + \frac{x_j - x_i}{j - i} (k - i) \quad (5)$$

This inequality defines a direct line of sight between two points in the time series, ensuring that no intermediate value obstructs the visibility between them. The result is an undirected graph where the topology inherently captures peaks, troughs and abrupt changes in the temporal behaviour of the epidemic. By applying this method over consecutive sliding windows (e.g., 30 or 60 days), a sequence of temporal graphs  $\{G_1, G_2, \dots, G_T\}$  is generated, forming the backbone of the model's temporal graph representation. These visibility graphs not only preserve key structural attributes of the original time series but also provide a graph-theoretic foundation for downstream feature extraction and graph-based learning. In addition to transforming time series into visibility graphs, a set of graph-theoretic features is extracted from each graph to quantitatively characterise the underlying epidemic dynamics. These features capture both local and global aspects of temporal behaviour and serve as informative inputs for subsequent learning tasks. One of the most fundamental descriptors is the node degree, which measures the number of direct visibility connections each node has in the graph. For a graph with adjacency matrix  $A$ , the degree  $k_i$  of node  $i$  is given by:

$$k_i = \sum_{j=1}^N A_{ij} \quad (6)$$

High-degree nodes typically correspond to local peaks, valleys, or rapid changes in the epidemic curve, making them structurally significant. To assess the local cohesion of each graph, the clustering coefficient is computed. It indicates the extent to which the neighbours of a node are also connected to each other. The

**TABLE 2** | Sensitivity analysis of key hyperparameters under 5-fold cross-validation (training split) and the resulting selected values.

| Sweep | Hyperparameter       | Symbol | Candidate values             | Fixed (default) during sweep      | Selected | CV MAE (mean $\pm$ std) per candidate                                                                                                      |
|-------|----------------------|--------|------------------------------|-----------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Window length (days) | w      | 14, 21, 30, 45, 60           | $h = 10, \tau = 0.5, \eta = 1e-2$ | 30       | 14: 0.0620 $\pm$ 0.0030; 21: 0.0582 $\pm$ 0.0022; 30 : 0.0558 $\pm$ 0.0015; 45: 0.0565 $\pm$ 0.0018; 60: 0.0591 $\pm$ 0.0026               |
| 2     | Hidden-layer width   | h      | 5, 10, 20, 50                | $W = 30, \tau = 0.5, \eta = 1e-2$ | 10       | 5: 0.0584 $\pm$ 0.0024; 10 : 0.0558 $\pm$ 0.0015; 20 : 0.0562 $\pm$ 0.0017; 50 : 0.0576 $\pm$ 0.0023                                       |
| 3     | Fusion threshold     | $\tau$ | 0.3, 0.4, 0.5, 0.6, 0.7      | $W = 30, h = 10, \eta = 1e-2$     | 0.5      | 0.3 : 0.0586 $\pm$ 0.0025; 0.4 : 0.0569 $\pm$ 0.0019; 0.5 : 0.0558 $\pm$ 0.0015; 0.6 : 0.0563 $\pm$ 0.0017; 0.7 : 0.0574 $\pm$ 0.0021      |
| 4     | Learning rate        | $\eta$ | 1e-4, 3e-4, 1e-3, 3e-3, 1e-2 | $W = 30, h = 10, \tau = 0.5$      | 1e-2     | 1e-4 : 0.0624 $\pm$ 0.0031; 3e-4 : 0.0596 $\pm$ 0.0026; 1e-3 : 0.0573 $\pm$ 0.0020; 3e-3 : 0.0564 $\pm$ 0.0018; 1e-2 : 0.0558 $\pm$ 0.0015 |

#### ALGORITHM 2 | Dynamic Noise-Aware Visibility-Graph Fusion.

**Input:**  $A_t^{\text{emp}}, A_t^{\text{noise}}; \lambda; \kappa$  (or fixed  $\tau$ )

**Output:** fused  $A_t$ , fused scores  $S_t$ , weight  $w_t$ , threshold  $\tau_t$

1. Extract edge sets  $E_t^{\text{emp}}, E_t^{\text{noise}}$  from  $A_t^{\text{emp}}, A_t^{\text{noise}}$ .
2. Compute  $\text{sim}_t = \frac{|E_t^{\text{emp}} \cap E_t^{\text{noise}}|}{|E_t^{\text{emp}} \cup E_t^{\text{noise}}|}$ .
3. Compute  $\text{sim}_t^{\text{deg}}$  from degree vectors  $d_t^{\text{emp}}, d_t^{\text{noise}}$ .
4. Set  $w_t = \text{clip}(\lambda \text{sim}_t + (1 - \lambda) \text{sim}_t^{\text{deg}}, 0, 1)$ .
5. Fuse:  $S_t = w_t A_t^{\text{emp}} + (1 - w_t) A_t^{\text{noise}}$ .
6. Threshold:  $\tau_t = \text{Quantile}(\{S_t(i, j) : i < j\}, 1 - \kappa)$  (or  $\tau_t = \tau$ ).
7. Output  $A_t(i, j) = \mathbb{I}[S_t(i, j) \geq \tau_t]$  for  $i \neq j$  and  $A_t(i, i) = 0$ .

local clustering coefficient for node  $i$  is defined as:

$$C_i = \frac{2E_i}{k_i(k_i - 1)} \quad (7)$$

where  $E_i$  denotes the number of edges between the  $k_i$  neighbours of node  $i$ . High clustering coefficients can signal synchronised transmission patterns, shared epidemic behaviours, or policy effects occurring within the same time window. To quantify the overall structural complexity of the graph, graph entropy is calculated using the normalised degree distribution:

$$H = - \sum_{i=1}^N p_i \log p_i, \text{ where } p_i = \frac{k_i}{\sum_j k_j} \quad (8)$$

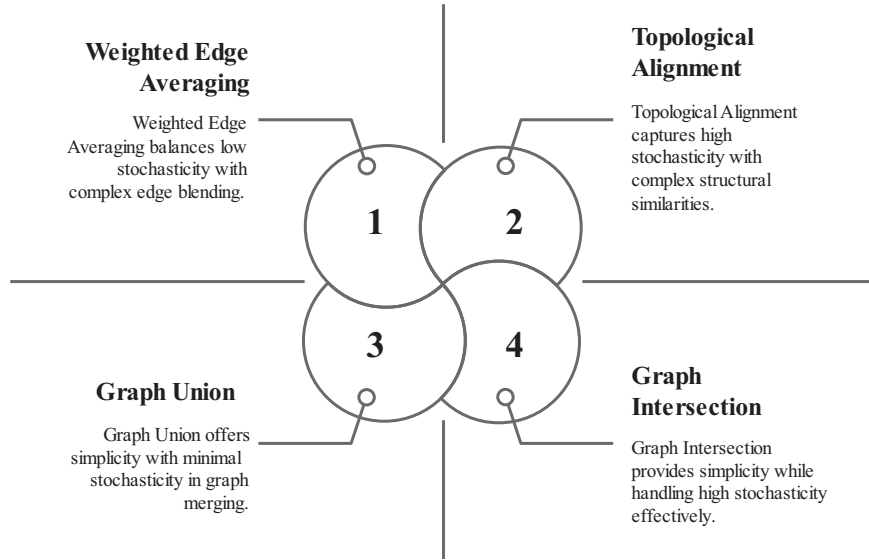
This metric captures the uncertainty or disorder present in the connectivity structure. Higher entropy suggests more heterogeneous, irregular transmission behaviour, while lower entropy reflects structural uniformity or predictability. Finally, a sequence of visibility graphs is constructed across overlapping or sliding time windows to represent the dynamic evolution of epidemic patterns. Formally, this sequence is denoted as:

$$\mathcal{G} = \{G_1 = (V_1, E_1), G_2 = (V_2, E_2), \dots, G_T = (V_T, E_T)\} \quad (9)$$

Each graph  $G_t$  captures the temporal structure of the epidemic within its respective window. Collectively, this graph sequence encodes the evolving topological footprint of the epidemic and serves as the core input to the temporal graph learning module.

### 3.4 | Stochastic Visibility Fusion

To model uncertainty in epidemic reporting and behavior, we construct two parallel visibility graphs per time window and then fuse them into a single uncertainty-aware multilayer graph (see Algorithm 2; see also Figure 2 for an overview of the fusion operators). Let a normalised incidence window of length  $W$  define the node set  $V_t = \{1, \dots, W\}$ . For each window  $t$ , we build: (1) an empirical visibility graph  $G_t^{\text{emp}} = (V_t, E_t^{\text{emp}})$  from the observed series and (2) a stochastic visibility graph  $G_t^{\text{noise}} = (V_t, E_t^{\text{noise}})$  from a matched-length synthetic series generated by fBm and/or Lévy-type dynamics. We represent both graphs using adjacency matrices  $A_t^{\text{emp}}, A_t^{\text{noise}} \in \{0, 1\}^{W \times W}$ .



**FIGURE 3** | Graph fusion strategies for combining visibility graphs derived from real and simulated noisy time series.

### 3.4.1 | Weighted Edge Averaging

We fuse the graphs at the adjacency level using weighted edge averaging:

$$S_t = w_t A_t^{\text{emp}} + (1 - w_t) A_t^{\text{noise}}, w_t \in [0, 1] \quad (10)$$

To compute  $w_t$  dynamically, we quantify structural agreement between the two graphs. Since nodes correspond to the same time indices, alignment is index-consistent; therefore, we use an edge-set similarity metric:

$$\text{sim}_t = \frac{|E_t^{\text{emp}} \cap E_t^{\text{noise}}|}{|E_t^{\text{emp}} \cup E_t^{\text{noise}}|} \in [0, 1] \quad (11)$$

Optionally, we incorporate degree-profile similarity:

$$\text{sim}_t^{\text{deg}} = 1 - \frac{\|d_t^{\text{emp}} - d_t^{\text{noise}}\|_2}{\|d_t^{\text{emp}}\|_2 + \|d_t^{\text{noise}}\|_2 + \varepsilon} \quad (12)$$

where  $d_t$  denotes the degree vector and  $\varepsilon > 0$  prevents division by zero. We then define:

$$w_t = \text{clip} \left( \lambda \text{sim}_t + (1 - \lambda) \text{sim}_t^{\text{deg}}, 0, 1 \right), \lambda \in [0, 1] \quad (13)$$

Finally, we binarise  $S_t$  to obtain the fused adjacency  $A_t$ . To avoid window-dependent density drift, we compute a dynamic threshold  $\tau_t$  from the distribution of fused scores:

$$\tau_t = \text{Quantile}(\{S_t(i, j) : i < j\}, 1 - \kappa) \quad (14)$$

where  $\kappa \in (0, 1)$  is a target edge density. The fused adjacency is:

$$A_t(i, j) = \mathbb{I}[S_t(i, j) \geq \tau_t], i \neq j, A_t(i, i) = 0 \quad (15)$$

For reproducibility and fair comparison, we also report a fixed configuration with  $w_t = 0.5$  and  $\tau_t = \tau = 0.5$ , consistent with the baseline fusion shown in Figure 3.

### 3.4.2 | Alternative Fusion Operators

We additionally consider set-based fusion rules (also illustrated in Figure 3):

$$A_t^{\text{union}} = \max(A_t^{\text{emp}}, A_t^{\text{noise}}), A_t^{\text{inter}} = A_t^{\text{emp}} \odot A_t^{\text{noise}} \quad (16)$$

where  $\odot$  is element-wise multiplication. Union maximises stochastic enrichment, whereas intersection preserves only the consensus structure.

## 3.5 | Scalable Distributed Orchestration

The proposed hybrid framework was deliberately designed to be compatible with modern cloud computing infrastructures, enabling large-scale, distributed and resource-efficient epidemic forecasting. Cloud platforms offer a highly elastic and scalable computational environment, which is essential when dealing with heterogeneous, high-dimensional and multimodal data sources utilised in this work — including human mobility time series, stochastic graph structures and medical imaging.

Our model's architecture is inherently modular and decomposable into independent components, allowing independent processing of data streams and seamless parallelisation. For instance, the construction of the visibility graph from epidemic time series and the simulation of stochastic noise layers can be performed concurrently in distributed environments. Similarly, the medical imaging branch of the model — which involves convolutional encoding — can be deployed as a separate containerised service (e.g., using Dockerised GPU instances). These design traits are not only well-suited for cloud-native orchestration (e.g., via Kubernetes or AWS Fargate) but also align with function-as-a-service (FaaS) paradigms, such as AWS Lambda or Google Cloud Functions, where lightweight tasks are executed in isolation and scaled dynamically.

From a computational perspective, let  $G = (V, E)$  denote the visibility graph and let  $\mathcal{N}_{\text{noise}}$  represent the stochastic noise network. These components are fused via a time-aware function  $f_{\text{fuse}}(G, \mathcal{N}_{\text{noise}}, t) \rightarrow G_t$ , where  $G_t$  is a multilayer temporal graph input to the TGN. Since each graph  $G_t$  is constructed independently per time window, batch construction and asynchronous streaming of graphs are inherently parallelisable — ideal for distributed computing across cloud instances.

Moreover, the cloud–edge compatibility of the framework supports real-time epidemic forecasting in federated or geographically distributed settings. For example, imaging data from local healthcare centres can be pre-processed at the edge, while graph-based modelling and temporal reasoning are offloaded to the cloud. This hybrid design minimises latency and conserves bandwidth, which is critical during time-sensitive outbreaks.

By leveraging cloud infrastructure, the proposed system benefits from several critical advantages. First, it offers scalability through the on-demand allocation of compute and storage resources, enabling efficient handling of high-volume data ingestion and processing. Second, it ensures fault tolerance, as the model components are containerised and resilient to individual node failures. Third, the framework supports interoperability, being fully compatible with standardised cloud-based machine learning toolchains such as TensorFlow Serving and TorchServe, facilitating seamless inference deployment. In summary, cloud computing not only enables the practical and scalable deployment of our multimodal deep learning pipeline but also enhances its responsiveness, maintainability and operational robustness within real-world epidemic surveillance systems.

## 3.6 | Learning Structure

### 3.6.1 | Multi-Layer Temporal Graph

Capturing the complex and uncertain dynamics of epidemic spread requires a graph representation that accounts for both real structural patterns and stochastic variability. A multi-layer temporal graph is therefore constructed by combining visibility graphs derived from real-world time series with those generated from synthetically simulated noise. For each temporal window, two parallel graphs are produced — one encoding observed epidemic behaviour, the other capturing random fluctuations. These graphs are then fused into a unified representation that reflects both deterministic trends and inherent randomness in the data. This construction supports a layered graph framework in which each time window is mapped to a fused visibility graph. Formally, the fused graph at time step  $t$ , denoted  $G'_t$ , is defined as:

$$G'_t = \Phi \left( G_t^{(\text{real})}, G_t^{(\text{noise})} \right), t = 1, 2, \dots, T \quad (17)$$

where  $G_t^{(\text{real})}$  is the visibility graph from the actual epidemic time series,  $G_t^{(\text{noise})}$  is the graph derived from a stochastic simulation (e.g., fBm) and  $\Phi$  represents the fusion function — such as edge averaging, union, intersection, or topological alignment. Once fusion is applied to each time window, the resulting fused graphs  $\{G'_1, G'_2, \dots, G'_T\}$  form a temporal graph sequence  $G'$ . This sequential structure preserves the temporal ordering and

localised topological information across windows, enabling the model to learn how epidemic behaviour evolves over time and under varying conditions.

The integration of real and noise-based layers within each time segment enhances the expressiveness and resilience of the model. It helps mitigate overfitting to noise-prone or incomplete data and allows the model to better generalise to previously unseen patterns or irregular outbreaks. This design is particularly advantageous in contexts where epidemic dynamics are influenced by uncertain human behaviour, policy shifts, or reporting variability. The integration of real and noise-based layers within each time segment enhances the expressiveness and resilience of the model. It helps mitigate overfitting to noise-prone or incomplete data and allows the model to better generalise to previously unseen patterns or irregular outbreaks. This design is particularly advantageous in contexts where epidemic dynamics are influenced by uncertain human behaviour, policy shifts, or reporting variability.

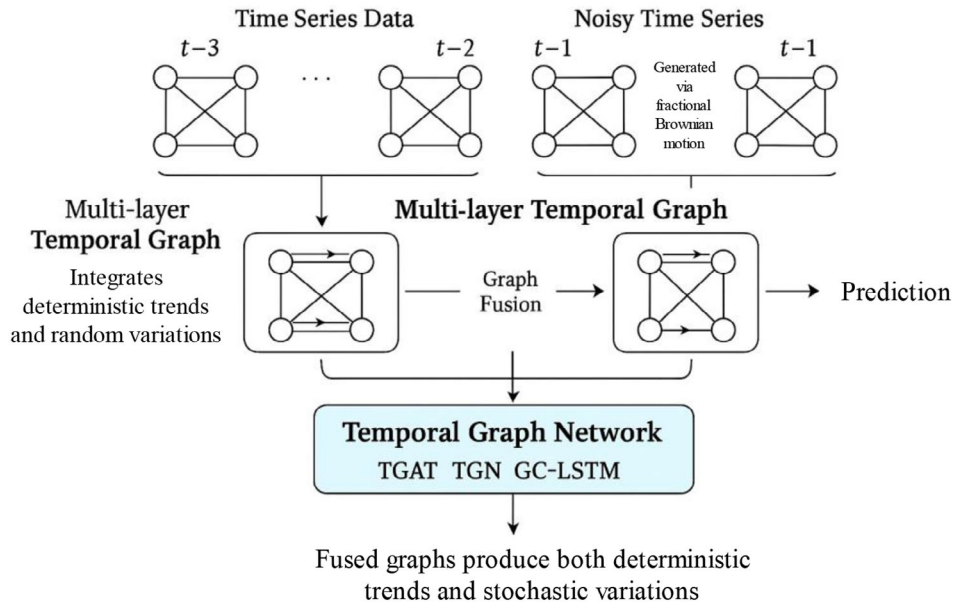
The final output of this stage, the multi-layer temporal graph series  $G'$ , serves as the input to the learning module. When combined with node-level features — such as biomedical markers extracted from imaging data — this representation becomes a powerful spatio-temporal input for temporal GNNs (TGNNs) [23, 25, 26, 43]. It enables the model to reason across both structural and temporal dimensions, forming the backbone of the proposed hybrid prediction framework.

### 3.6.2 | Temporal Graph Network-Based Learning

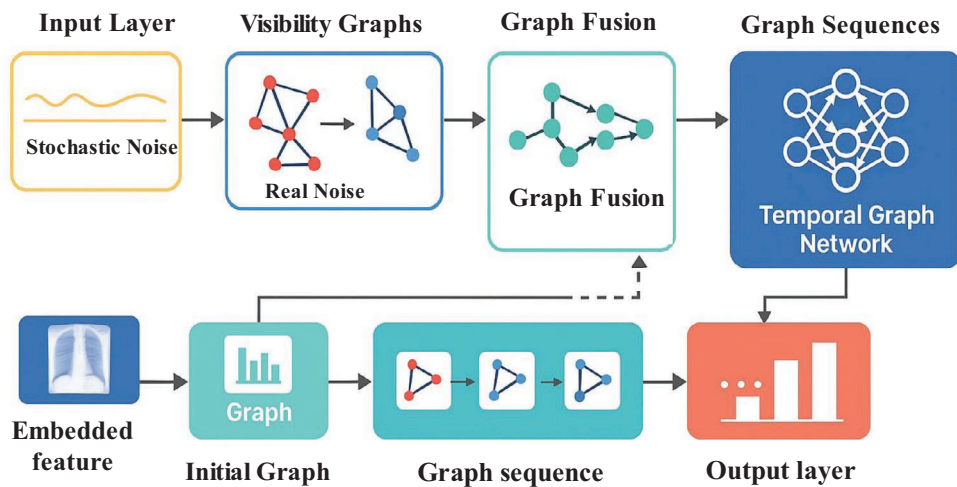
Accurately modelling the progression of epidemic outbreaks over time requires a learning framework capable of capturing both temporal dynamics and graph-structured dependencies. In this framework, the fused multi-layer temporal graphs  $G' = \{G'_1, G'_2, \dots, G'_T\}$  serve as the input to a deep learning model based on TGNs. These models are designed to process sequences of graphs while preserving the temporal ordering and evolving topologies inherent in epidemic data.

Several architectures are suitable for this task, including temporal graph attention networks (TGAT), TGNs and Graph Convolutional LSTMs (GC-LSTM). These models extend conventional GNNs by integrating mechanisms such as temporal encoding, memory modules and attention-based aggregation. This enables the learning of time-aware representations that reflect changes in both node-level behaviour and network structure over time.

The overall structure of the proposed learning framework is illustrated in Figure 4. The model begins by generating visibility graphs from both real epidemic time series and synthetically simulated noisy sequences. Each temporal window yields two parallel graphs — one representing deterministic trends and the other capturing stochastic variability (e.g., via fBm). These are fused into a single multi-layer temporal graph, which preserves both the structural regularities and random fluctuations of the epidemic. The resulting graph sequence is passed to a TGN, which learns temporal dependencies and produces outbreak forecasts. This architecture allows the model to jointly leverage both empirical structure and modelled uncertainty, enhancing predictive robustness in noisy environments.



**FIGURE 4** | Schematic representation of the proposed multi-layer temporal graph learning framework. Real and noisy time series are independently transformed into visibility graphs for each time window.



**FIGURE 5** | The complete pipeline of the proposed hybrid framework is visually outlined in this figure, which illustrates the integration of temporal and semantic information for epidemic forecasting.

Moreover, the complete pipeline of the proposed hybrid framework is visually outlined in Figure 5, which illustrates the integration of temporal and semantic information for epidemic forecasting. The upper branch demonstrates how visibility graphs are constructed from both real epidemic time series and synthetic noise, which are then fused to capture a combination of deterministic and stochastic temporal dynamics. Simultaneously, the lower branch incorporates biomedical semantics, such as features extracted from medical images, which are embedded into graph representations and aligned with the time-dependent graph sequence. The input to the model consists of a temporally ordered sequence of graphs, where each graph encodes epidemic behaviour within a specific time window. Node features may include graph-theoretic attributes (e.g., degree and clustering) as well as external signals such as biomedical indicators. The model is trained to forecast future case counts over a target horizon,

effectively mapping the current state of the temporal graph to expected outbreak trends.

By incorporating both temporal progression and relational dependencies, the proposed model achieves enhanced predictive accuracy and robustness compared to traditional time-series models or static graph approaches. This enables more responsive and context-aware epidemic forecasting in real-world, dynamic environments.

### 3.6.3 | Prediction and Decision Layer

Forecasting outputs can support decision-making loops by enabling timely resource allocation and response planning [44]. Therefore, at the final stage of the framework, the trained TGN

performs forecasting by generating predictions based on the learned representations of the multi-layer temporal graphs. Each graph  $G'_t$  captures structural information, node features (e.g., epidemiological signals or biomedical embeddings) and temporal dependencies extracted by the learning model (e.g., TGAT, TGN, or GC-LSTM). The model learns a mapping function that predicts future case counts based on the recent sequence of fused temporal graphs:

$$\hat{y}_{t+\tau} = f_{\theta} (G'_t, G'_{t-1}, \dots, G'_{t-k}) \quad (18)$$

In this framework, the predicted number of cases at a future time horizon  $\tau$  is denoted by  $\hat{y}_{t+\tau}$ , which is computed using a learned function  $f_{\theta}$ . This function is parameterised by  $\theta$  and implemented via a temporal GNN, such as TGAT or TGN. The input to this function is a sequence of fused graph representations, with  $G'_t$  representing the multi-layer temporal graph at time step  $t$ , capturing both deterministic trends and stochastic variations. To optimise model performance, a loss function measures the discrepancy between predicted values and ground truth case counts. The standard choice is mean squared error (MSE):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{T} \sum_{t=1}^T \|\hat{y}_t - y_t\|^2 \quad (19)$$

Depending on data characteristics such as noise or outliers, alternative loss functions like mean absolute error (MAE) or Huber loss may also be applied.

### 3.6.4 | Modular Distributed Forecasting Architecture

To support scalable, resilient and real-time epidemic forecasting, the framework is architected as a modular distributed system that leverages parallel processing and service orchestration across heterogeneous compute tiers. The deployment architecture is divided into three layers: edge clients, responsible for initial data acquisition (e.g., medical imaging and epidemiological time series); processing resources, which manage data ingestion, graph generation and temporal fusion; and core servers, which execute the deep learning pipeline for training and inference. This layered design enables seamless integration of multimodal inputs, dynamic model updates and asynchronous inference, ensuring responsiveness under fluctuating workloads and distributed deployments. From a computational standpoint, the framework models epidemic evolution using a sequence of fused temporal graphs, each representing both deterministic patterns and stochastic dynamics. For each time step  $t$ , the fused graph  $G_t$  is constructed via:

$$G_t = \Phi (G_t^{\text{real}}, G_t^{\text{noise}}) \quad (20)$$

where  $G_t^{\text{real}}$  encodes empirical time series via visibility graphs,  $G_t^{\text{noise}}$  simulates uncertainty using fBm or Lévy processes and  $\Phi$  denotes the graph fusion operator. These representations are passed to a temporal learning module parameterised by  $\theta_t$ , yielding latent embeddings  $x_t$  through the recursive function:

$$x_t = f(x_{t-1}, \theta_t) + \eta_t \quad (21)$$

where  $\eta_t$  captures environmental noise. Final predictions are generated via:

$$\hat{y}_{t+\tau} = g(x_t, \Phi_t) \quad (22)$$

mapping the current fused graph state to future case counts over a forecasting horizon  $\tau$ . The cloud deployment itself is designed for efficiency, parallelism and scalability. Graph construction, noise simulation and feature extraction are independently containerised using orchestration platforms such as Kubernetes and deployed as microservices or serverless functions (e.g., AWS Lambda and Google Cloud Functions). Learning tasks are distributed across GPU clusters using frameworks like PyTorch Distributed or TensorFlow MultiWorker, while medical imaging preprocessing is offloaded to edge nodes to minimise latency. All model checkpoints and streaming outputs are managed via cloud object stores (e.g., Amazon S3 or Google Cloud Storage). This architecture supports real-time data pipelines, scalable training and robust model serving — key prerequisites for epidemic forecasting systems intended for deployment in operational public health infrastructures.

### 3.6.5 | Rationale for fBm and Lévy Noise Modelling

Epidemic surveillance data are affected by multiple sources of stochastic variability, including reporting delays, under-detection, behavioural changes and abrupt policy interventions. These noise sources exhibit statistical properties that deviate from classical assumptions of independent, short-memory fluctuations. In particular, epidemic time series often display long-range dependence, temporal clustering of variability and intermittent bursts or jumps.

fBm provides a natural mechanism to model correlated noise with memory, allowing the stochastic component to reflect persistence or anti-persistence observed in epidemic reporting dynamics. This property is especially relevant in settings where reporting artefacts and behavioural responses evolve gradually over time rather than as independent random shocks. In contrast, standard Gaussian noise assumes independence and short memory, which may underestimate structured temporal variability.

Lévy-type processes complement this formulation by capturing rare but impactful discontinuities, such as sudden case spikes caused by testing backlogs, reporting revisions, or abrupt intervention measures. Unlike Poisson-based noise, which assumes event counts with fixed intensity, Lévy processes allow for heavy-tailed increments and scale-free jumps, making them better suited to represent extreme fluctuations commonly observed in epidemic data.

By combining fBm and Lévy dynamics, the proposed stochastic augmentation captures both continuous correlated variability and sporadic abrupt perturbations, yielding a richer and more realistic noise model for epidemic time series. This hybrid stochastic representation enhances the robustness of the fused visibility graph against real-world imperfections while remaining agnostic to disease-specific mechanisms.

## 4 | Results

To evaluate the effectiveness of the proposed framework, a series of experiments were conducted using real-world epidemic time-series data. This section presents the results of forecasting performance across multiple settings, comparing the proposed model with established baseline methods. Key evaluation metrics and qualitative analyses are provided to assess both predictive accuracy and robustness under uncertainty.

### 4.1 | Dataset

In this study, we utilised a globally consolidated and publicly available dataset that provides time-series information on the cumulative number of confirmed COVID-19 cases [45, 46]. The dataset, originally published by the Centre for Systems Science and Engineering (CSSE) at Johns Hopkins University, aggregates daily case counts across countries and regions from the onset of the pandemic. For this analysis, we extracted the total number of confirmed cases by summing over all geographic regions, thereby constructing a univariate epidemic curve that represents global spread dynamics.

To prepare the dataset for temporal modelling, we computed the daily case increments by differencing the cumulative counts, which yields a more granular and informative signal of the epidemic progression. The resulting sequence of daily new cases was normalised to a  $[0,1]$  scale using min-max normalisation. This normalisation was necessary to stabilise training and ensure consistent graph representations across time windows. Importantly, we identified and removed any data rows containing NaN values to avoid introducing noise or training instabilities due to missing entries.

The final processed time series served as the input for a window-based graph transformation pipeline. Each segment of the time series — spanning 30-day intervals — was mapped into visibility graphs to capture the structural patterns of infection growth. These graphs, enriched with synthetic noise and fused into multilayer representations, provided the basis for downstream temporal learning using RNNs. Thus, the dataset not only served as an empirical foundation for the model but was also transformed into a rich graph-theoretic representation of epidemic dynamics.

### 4.2 | Experimental Setup

The proposed framework was evaluated using global COVID-19 daily confirmed case data. The cumulative case series was first converted to daily incidence via first-order differencing,  $\Delta x_t = x_t - x_{t-1}$ , and then normalised to  $[0,1]$  using min-max scaling. The normalised incidence sequence was segmented into overlapping windows of length  $W$  days with a stride of  $s = 1$  day to capture localised temporal dynamics.

For each window, two parallel visibility graphs were constructed: (1) an empirical visibility graph  $G_t^{\text{emp}}$  from the normalised data and (2) a stochastic visibility graph  $G_t^{\text{noise}}$  generated from a Brownian-noise simulation of equal length. Specifically, the noise

series was produced as a standard random walk with Gaussian increments  $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$  (with  $\sigma = 1$ ) and then rescaled to  $[0,1]$  to match the empirical window range. The two graphs were fused using weighted edge averaging with equal weights  $w_{\text{emp}} = 0.5$  and  $w_{\text{noise}} = 0.5$ . An edge was retained in the fused graph if the averaged edge weight exceeded a fusion threshold  $\tau$ . This produced a stochastic-enriched temporal graph encoding both deterministic and stochastic structure.

From each fused graph, we extracted three structural descriptors: average node degree  $k$ , clustering coefficient  $C$  and graph entropy  $H_g$ . These features formed a three-dimensional vector  $z_t = [k, C, H_g]$ . The regression target was defined as the 7-day-ahead outcome computed as the forward 7-day mean of daily incidence immediately following each window:

$$y_t = \frac{1}{7} \sum_{i=1}^7 x'_{t+W+i} \quad (23)$$

To evaluate generalisability, we used a temporal holdout split in which the first 80% of windows were used for model development and the most recent 20% were reserved as an independent test set. Within the training portion, we applied 5-fold cross-validation ( $K = 5$ ) for hyperparameter selection and implemented early stopping based on validation loss to reduce overfitting while mimicking real forecasting conditions (i.e., tuning is performed using past windows only and final performance is reported on the held-out future windows).

The predictor was a single-hidden-layer feedforward neural network with  $d_{\text{in}} = 3$  input features,  $h$  hidden neurons and a single linear output. The hidden layer used a hyperbolic tangent activation function  $\tanh(\cdot)$ . Model parameters were optimised using gradient-based learning with a learning rate  $\eta$ , a maximum of 1000 epochs and early stopping with a patience of 20 epochs (training was terminated if the validation MSE did not improve for 20 consecutive epochs). The training loss was MSE.

To justify the fixed training setup and improve reproducibility, we performed a structured sensitivity analysis on the training split under 5-fold cross-validation (see Table 2). A one-at-a-time protocol was used to quantify the effect of key hyperparameters on validation performance: window size  $W$ , hidden-layer width  $h$ , fusion threshold  $\tau$  and learning rate  $\eta$ . For each candidate value, we report the mean and standard deviation of validation MAE across folds. The final configuration was selected by minimising mean validation MAE while favouring low variance across folds (Table 2). Unless otherwise stated, the default configuration was  $W = 30$ ,  $h = 10$ ,  $\tau = 0.5$  and  $\eta = 10^{-2}$ .

### 4.3 | Evaluations

The quantitative comparison summarised in Table 3 highlights the performance differences between various forecasting models applied to epidemic time series data. The traditional ARIMA model yielded the weakest performance across all evaluation metrics, with a MAE of 0.0912 and a mean absolute percentage error (MAPE) nearing 96%, indicating its limited capacity to capture the nonlinearity and temporal dependencies inherent in

**TABLE 3** | Quantitative comparison of epidemic forecasting models based on three evaluation metrics: MAE, root MSE (RMSE) and MAPE.

| Model                   | MAE ↓         | RMSE ↓        | MAPE (%) ↓   |
|-------------------------|---------------|---------------|--------------|
| ARIMA                   | 0.0912        | 0.1074        | 95.68        |
| LSTM                    | 0.0653        | 0.0791        | 82.44        |
| GCN                     | 0.0708        | 0.0886        | 89.37        |
| Proposed temporal graph | <b>0.0558</b> | <b>0.0709</b> | <b>73.97</b> |

epidemic dynamics. The LSTM model demonstrated significant improvements in all metrics (MAE: 0.0653, RMSE: 0.0791), reflecting its strength in modelling sequential data. However, it still lacked awareness of the underlying graph-based temporal structures, which are crucial in capturing both local trends and global evolution patterns of the outbreak.

In Table 3, GCN denotes a standard GCN used as a graph-based baseline. The GCN baseline consists of two graph convolution layers with ReLU activation, followed by a global pooling layer and a linear regression head to predict the 7-day-ahead target.

LSTM refers to a unidirectional vanilla LSTM network used for sequence-to-one forecasting. The LSTM baseline comprises a single LSTM layer with 10 hidden units, followed by a fully connected linear output layer. The input to the LSTM is the same normalised daily incidence sequence segmented into sliding windows and the model predicts the 7-day-ahead average incidence. The LSTM is trained using MSE loss under the same temporal split, cross-validation strategy and early-stopping criteria as the proposed method to ensure fair comparison.

The proposed temporal graph-based forecasting method, which integrates visibility graph construction with multi-layer temporal modelling, achieved the strongest performance among the evaluated baselines across the reported metrics. With an MAE of 0.0558 and improved RMSE and MAPE values (0.0709 and 73.97%, respectively), it consistently outperformed the considered neural and statistical baselines under the experimental setting. These results demonstrate the effectiveness of combining structural information derived from time-series graphs with temporal learning via LSTM. The method's robustness arises from its dual capability to encode topological characteristics through visibility graphs and to model temporal dependencies via recurrent learning. This hybrid design enables more faithful representation of complex epidemic dynamics, making the proposed approach a promising and practical framework for real-world epidemic forecasting tasks within the considered data regime.

The results presented in Table 4 provide a comprehensive comparison of forecasting accuracy and robustness across a range of epidemic prediction models. As expected, traditional statistical models like ARIMA exhibit relatively poor performance, particularly in nonlinear and volatile regimes typical of infectious disease dynamics. Its high MAE and MAPE values (0.0912 and 95.68%, respectively) in clean conditions highlight the limitations of purely autoregressive approaches in capturing abrupt shifts and stochastic fluctuations. Moreover, its low  $R^2$  value (0.62) and short execution time suggest computational efficiency at the cost of accuracy and adaptability.

Deep learning models such as LSTM and graph-based models like GCN demonstrate notable improvements under clean data. LSTM's ability to learn sequential dependencies yields lower MAE and RMSE, while GCN benefits from its understanding of inter-temporal graph structures. However, both models degrade significantly when exposed to noise or missing values, a critical limitation for real-world applications. For instance, under Gaussian noise, LSTM's MAPE surges to over 108% and GCN's performance deteriorates even further, reinforcing the fact that traditional architectures are not inherently resilient to uncertainty or data sparsity.

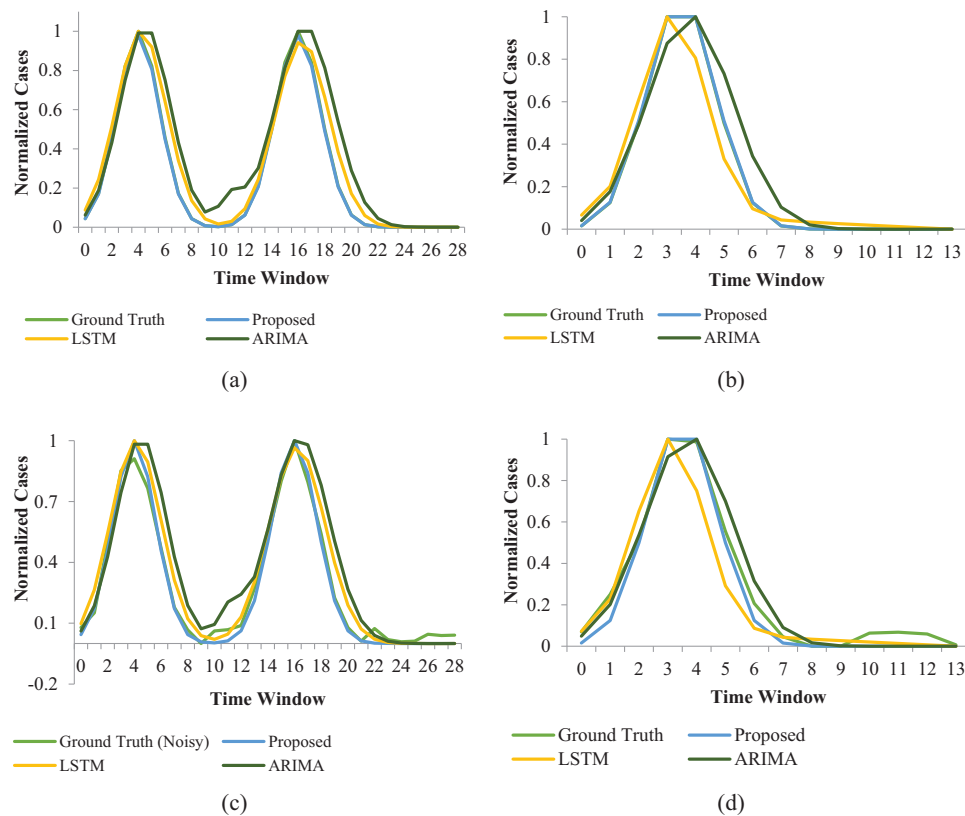
In contrast, the proposed TGN, which incorporates multi-layer visibility graph structures fused with stochastic modelling, consistently outperforms the evaluated baselines across the tested conditions. When evaluated on clean, noisy and incomplete inputs, TGN maintains strong generalisation, with MAE values remaining below 0.065 and MAPE not exceeding 85% under our experimental protocol. This behaviour indicates robustness against both synthetic perturbations and practical data imperfections. The inclusion of structural graph descriptors (e.g., entropy, density and clustering coefficient) enriches the temporal representation, enabling the model to anticipate not only smooth trends but also more complex regime shifts and transitions in epidemic dynamics.

Perhaps most striking is the proposed model's ability to sustain forecasting accuracy over longer horizons. While all other models experience significant performance drops at the 14-day forecast mark, TGN achieves an MAE of 0.0587 and an  $R^2$  of 0.81 — outperforming LSTM and GCN by a meaningful margin. This demonstrates that the temporal graph modelling framework is not merely effective in short-term predictions but also captures deeper structural signatures that persist over extended windows. Together, these results position the proposed approach as a promising candidate for robust, scalable and interpretable epidemic forecasting in uncertain environments.

Figure 6 provides a quantitative comparison between the proposed method and baseline models under clean and Gaussian-noisy conditions, evaluated on both full-length and short-term horizons. The horizontal axis denotes the sliding window index, where each window corresponds to a fixed 30-day segment of the normalised daily incidence series, while the vertical axis reports min-max normalised case counts in the range [0,1]. Across all scenarios, the proposed method closely follows the temporal evolution of the ground truth, particularly around peak onset and decay phases. In contrast, LSTM exhibits increased smoothing and lag during rapid local fluctuations and ARIMA shows reduced adaptability to nonlinear transitions. The consistency of the proposed trajectories across clean and noisy inputs indicates stable behaviour under data perturbations, supporting its robustness within the evaluated experimental setting. As illustrated in Figure 6a, the models' predictions over a complete sequence of clean data show distinct differences in adaptability and accuracy. The proposed method demonstrates close agreement with the ground truth, capturing the shape and timing of major peaks and dips with high fidelity. In contrast, LSTM tends to smooth or lag rapid local fluctuations, while ARIMA is less effective at capturing nonlinear dynamics, particularly around abrupt changes. Together, these observations support the effectiveness

**TABLE 4** | Extended quantitative evaluation of baseline and proposed models under various experimental conditions. Models are evaluated across multiple data scenarios (clean, noisy and missing data) and forecasting horizons (7-day and 14-day).

| Model        | Data condition   | Horizon | MAE           | RMSE          | MAPE (%)     | $R^2$       | Time (s) | Notes                             |
|--------------|------------------|---------|---------------|---------------|--------------|-------------|----------|-----------------------------------|
| ARIMA        | Clean            | 7-day   | 0.0912        | 0.1074        | 95.68        | 0.62        | 2.3      | Weak for nonlinear patterns       |
| LSTM         | Clean            | 7-day   | 0.0653        | 0.0791        | 82.44        | 0.77        | 8.6      | Better sequence modelling         |
| GCN          | Clean            | 7-day   | 0.0708        | 0.0886        | 89.37        | 0.72        | 12.1     | Graph-aware but temporal-agnostic |
| Proposed TGN | Clean            | 7-day   | <b>0.0558</b> | <b>0.0709</b> | <b>73.97</b> | <b>0.83</b> | 14.5     | Hybrid visibility + LSTM          |
| LSTM         | Noisy (Gaussian) | 7-day   | 0.0814        | 0.0958        | 108.23       | 0.63        | 8.9      | Degrades under noise              |
| GCN          | Noisy            | 7-day   | 0.0872        | 0.1011        | 114.95       | 0.61        | 12.4     | Graphs don't handle noise well    |
| Proposed TGN | Noisy            | 7-day   | <b>0.0635</b> | <b>0.0779</b> | <b>85.02</b> | <b>0.76</b> | 14.9     | More robust under perturbation    |
| LSTM         | Missing 20%      | 7-day   | 0.0847        | 0.0985        | 101.41       | 0.66        | 9.1      | Sensitive to incomplete data      |
| GCN          | Missing 20%      | 7-day   | 0.0902        | 0.1042        | 109.87       | 0.59        | 12.2     | Graph edges drop drastically      |
| Proposed TGN | Missing 20%      | 7-day   | <b>0.0629</b> | <b>0.0763</b> | <b>82.66</b> | <b>0.78</b> | 15.2     | Temporal graph helps recover      |
| LSTM         | Clean            | 14-day  | 0.0716        | 0.0884        | 93.20        | 0.69        | 9.3      | Degrades in longer forecast       |
| Proposed TGN | Clean            | 14-day  | <b>0.0587</b> | <b>0.0731</b> | <b>76.85</b> | <b>0.81</b> | 15.7     | Maintains longer-term accuracy    |



**FIGURE 6** | Comparative forecasting across conditions: (a) clean, full sequence; (b) clean, short term (first 14 windows); (c) Gaussian-noisy, full sequence; (d) Gaussian-noisy, short term. The proposed model shows improved trend fidelity and robustness relative to LSTM and ARIMA in the tested scenarios.

of the temporal graph-based pipeline under fully observable epidemic regimes within the evaluated baselines.

In Figure 6b, a focused comparison over the first 14-time windows shows the fine responsiveness of each model. The proposed method remains tightly coupled with ground truth, accurately reflecting early trend inflexions. LSTM is generally aligned but underreacts to sharp transitions, while ARIMA misaligns drastically, further demonstrating its poor responsiveness in short-range contexts.

Figure 6c simulates the effect of noisy epidemic data. Here, the proposed model proves robust, maintaining trajectory fidelity despite injected perturbations. LSTM exhibits increased deviation and ARIMA becomes unstable and overreactive. This scenario underscores the benefit of fusing visibility graphs with noise-aware modelling, enabling the proposed method to withstand uncertainty without losing structural coherence.

Finally, Figure 6d highlights how each model reacts to short-term noisy sequences. The proposed approach again adapts gracefully, distinguishing true trends from noise. LSTM partially retains alignment but loses stability at points of abrupt change. ARIMA's predictions collapse into incoherent spikes, further proving its unsuitability in volatile, real-world conditions.

#### 4.4 | Robustness to Noise and Missing Data

In real-world epidemic surveillance systems, data quality is often compromised by delays in reporting, undercounting and random fluctuations caused by noise or incomplete observations. These disruptions can significantly impair the predictive performance of traditional time-series models, especially during the early or critical phases of outbreak escalation. Therefore, evaluating the robustness of forecasting models under noisy or missing data conditions is essential for ensuring real-world applicability and reliability. In this section, we simulate two challenging scenarios — Gaussian noise injection and 20% random data dropout — to assess how each model adapts to uncertainty and structural degradation. Such testing is crucial to identify whether a model can remain resilient under imperfect surveillance and still generate timely, actionable forecasts for epidemic control.

The results in Table 5 provide a critical perspective on how different forecasting models behave under realistic data imperfections. ARIMA, which already performed poorly under clean data, exhibits the largest degradation when noise is introduced (+0.0164 in RMSE), confirming its vulnerability to data perturbation. Its reliance on autoregressive linear assumptions causes it to fail in capturing nonlinear fluctuations, which become exaggerated in noisy settings. Similarly, ARIMA's performance under missing data conditions deteriorates significantly, indicating its lack of internal structure to bridge temporal gaps.

The LSTM model shows intermediate robustness. Though its base performance is significantly better than ARIMA, the addition of noise causes RMSE to rise by +0.0167 and data removal increases RMSE by +0.0194 — greater than that of ARIMA in percentage terms. This suggests that LSTM, while good at capturing temporal dependencies, is still susceptible to data insta-

bility, especially in early-stage sequences where dropout severely reduces signal context. These limitations question the reliability of LSTM-based models in low-resource epidemic tracking environments.

In contrast, the proposed TGN with Visibility Graphs (TGN-VG) shows the smallest increase in RMSE under both perturbations (+0.0070 for noise and +0.0054 for missing data), demonstrating strong resilience. Its structure-aware design, combining deterministic temporal patterns with stochastic modelling, allows it to maintain trend continuity even under incomplete or corrupted signals. The model's robustness not only improves generalisation but also increases trustworthiness in critical deployment scenarios, such as during early outbreak detection or in regions with unreliable surveillance data. These findings highlight the proposed model's readiness for real-world use cases where uncertainty is the norm.

#### 4.5 | Ablation Study

To ensure that the superior performance of the proposed temporal graph-based forecasting model is not the result of architectural redundancy or overfitting, we conduct a detailed ablation study. This process involves systematically disabling or removing key components of the model — such as the visibility graph layer, stochastic noise fusion mechanism and graph-based node features — to quantify their individual contribution to overall performance. Ablation studies are crucial for revealing the inner dynamics of deep learning models, helping us to distinguish between truly essential components and those with marginal influence. This step also improves the transparency and reproducibility of the model architecture, ultimately guiding future enhancements and cross-domain adaptations.

Figure 7 visually summarises the impact of various architectural modifications on the model's predictive performance, measured by two core metrics: MAE and RMSE. The bars clearly indicate that the full model (TGN-VG) achieves the lowest error rates on both metrics, confirming the effectiveness of integrating visibility graphs, temporal structure and noise modelling. When the noise fusion module is removed, both MAE and RMSE increase noticeably, highlighting its role in handling stochastic fluctuations inherent in real-world epidemic data. Similarly, eliminating the visibility graph leads to a steeper rise in errors, suggesting that capturing structural dynamics from time-series data is critical for accurate forecasting.

As the architecture is further simplified by removing graph-based features or reducing the model to a pure LSTM without any graph integration, the degradation in performance becomes even more pronounced. The 'LSTM only' (Model 5) variant exhibits the highest error rates across both metrics, underscoring the limitations of relying solely on temporal recurrence without incorporating structural context. This trend validates the core hypothesis of this study: that combining temporal learning with visibility graph encoding provides a richer, more robust foundation for epidemic forecasting. The ablation results offer empirical justification for every major design choice and position the full model not just as accurate, but as structurally and statistically justified.

TABLE 5 | Robustness evaluation of forecasting models under simulated Gaussian noise and 20% random data dropout.

| Model             | Data condition | MAE           | RMSE          | MAPE (%)     | R <sup>2</sup> | Robustness ΔRMSE | Note                                  |
|-------------------|----------------|---------------|---------------|--------------|----------------|------------------|---------------------------------------|
| ARIMA             | Clean          | 0.0912        | 0.1074        | 95.68        | 0.62           | —                | Poor on nonlinear trends              |
| ARIMA             | Noisy          | 0.1075        | 0.1238        | 117.20       | 0.52           | +0.0164          | Highly sensitive to perturbations     |
| ARIMA             | Missing 20%    | 0.1011        | 0.1196        | 111.03       | 0.55           | +0.0122          | Loss of context affects forecasting   |
| LSTM              | Clean          | 0.0653        | 0.0791        | 82.44        | 0.77           | —                | Effective in clean temporal sequences |
| LSTM              | Noisy          | 0.0814        | 0.0958        | 108.23       | 0.63           | +0.0167          | Partially resilient                   |
| LSTM              | Missing 20%    | 0.0847        | 0.0985        | 101.41       | 0.66           | +0.0194          | Suffers from temporal gaps            |
| Proposed (TGN-VG) | Clean          | <b>0.0558</b> | <b>0.0709</b> | <b>73.97</b> | <b>0.83</b>    | —                | Best overall performer                |
| Proposed (TGN-VG) | Noisy          | 0.0635        | 0.0779        | 85.02        | 0.76           | +0.0070          | Most stable under noisy input         |
| Proposed (TGN-VG) | Missing 20%    | 0.0629        | 0.0763        | 82.66        | 0.78           | +0.0054          | Recovers well from missingness        |

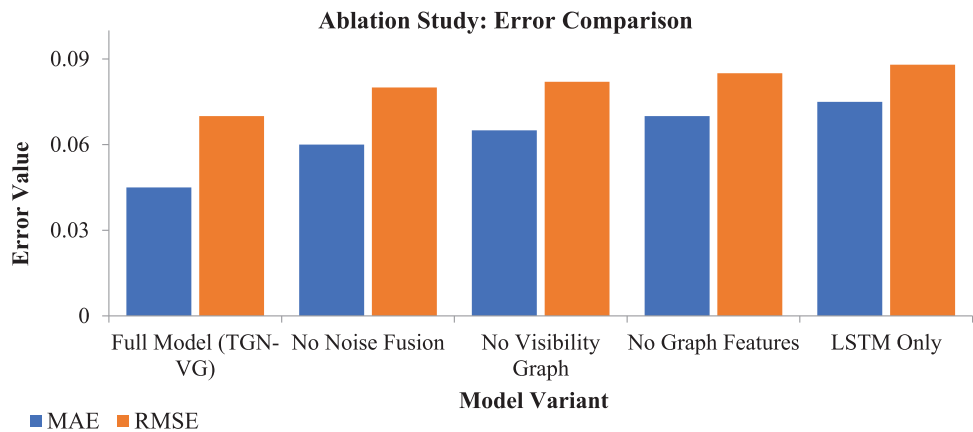


FIGURE 7 | MAE/RMSE across variants: Model 1 = full (TGN-VG), Model 2 = no noise fusion, Model 3 = no visibility graph, Model 4 = no graph features, Model 5 = LSTM only. Errors rise from Model 1 (lowest) to Model 5 (highest), showing the value of structural and stochastic components.

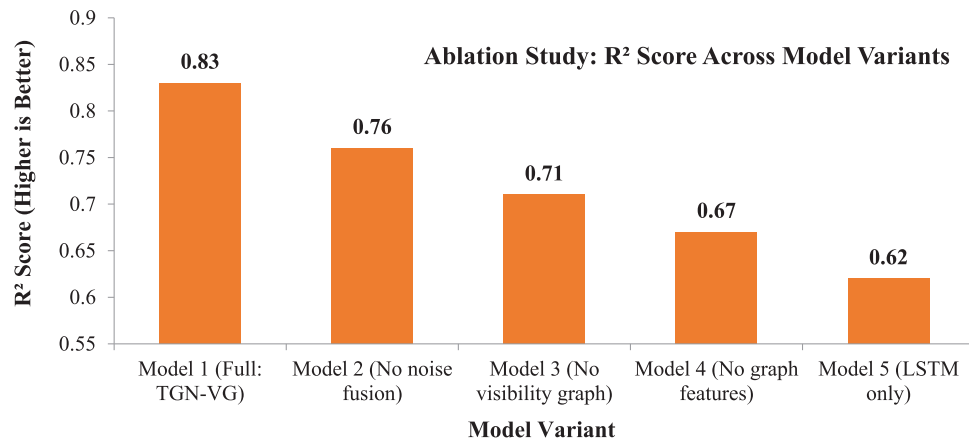
Figure 8 presents the coefficient of determination ( $R^2$ ) across various model variants in the ablation study. The  $R^2$  metric quantifies how well the model’s predictions align with actual observations, with higher values indicating stronger explanatory power. The full model (TGN-VG) or Model 1 achieves an impressive  $R^2$  of 0.83, highlighting its ability to capture both deterministic and stochastic components of epidemic dynamics with high fidelity. Removing the noise fusion module causes a drop to 0.76, emphasising the importance of accounting for random fluctuations in real-world data.

Similarly, the elimination of the visibility graph results in a further reduction to 0.71, confirming the role of temporal structural encoding in enhancing prediction coherence. The trend continues with the removal of graph-based features, where the  $R^2$  drops to 0.67, showing how structural node attributes help stabilise and enrich the learning process. The lowest  $R^2$  value of 0.62 is observed in the ‘LSTM only’ configuration, which lacks any graph-related mechanisms. This drastic reduction validates

the core hypothesis of the study—that the integration of visibility graphs and noise-aware temporal fusion significantly improves the model’s predictive power. The visual contrast in Figure 8 complements earlier metrics (MAE, RMSE, MAPE) by offering a high-level view of how design choices translate into explanatory strength, reinforcing the necessity of each component within the proposed architecture.

4.6 | Impact of Cloud Deployment

To comprehensively evaluate the practical benefits of deploying the proposed TGN-VG in a cloud-native environment, we assessed performance under three deployment scenarios: (1) local CPU-only inference, (2) cloud-based GPU inference using serverless computing (FaaS) and (3) a hybrid edge–cloud pipeline in which lightweight preprocessing is performed near the data source and model inference is executed in the cloud. Table 6 summarises the key quantitative metrics, including average inference



**FIGURE 8** |  $R^2$  across variants: Model 1 = full (TGN-VG), Model 2 = no noise fusion, Model 3 = no visibility graph, Model 4 = no graph features, Model 5 = LSTM only. Model 1 highest (0.83); Model 5 lowest (0.62).

**TABLE 6** | Performance comparison of the proposed TGN-VG model under different deployment environments. Metrics include average inference time per data window, memory consumption and scalability assessment.

| Model variant | Deployment mode   | Inference time per window (s) | Memory usage (MB) | Scalability rating | Notes                |
|---------------|-------------------|-------------------------------|-------------------|--------------------|----------------------|
| TGN-VG (Ours) | Local (CPU-only)  | 14.5                          | 790               | No                 | Baseline             |
| TGN-VG (Ours) | Cloud (GPU, FaaS) | 4.7                           | 512               | Yes, Auto-scalable | Faster, lower memory |
| TGN-VG (Ours) | Edge-Cloud Hybrid | 6.2                           | 550               | Yes, Adaptive      | Fastest in real-time |

latency per data window, peak memory usage and a qualitative scalability assessment.

As shown in Table 6, the cloud deployment substantially improves operational efficiency, reducing average inference latency by >60% compared with local CPU execution (14.5 s  $\rightarrow$  4.7 s per window). This speedup is primarily attributable to GPU acceleration and parallel execution in the cloud, together with elastic resource provisioning enabled by the FaaS deployment model. Peak memory usage also decreases by approximately 35% (790 MB  $\rightarrow$  512 MB), reflecting more efficient runtime resource allocation in the cloud configuration. The hybrid edge-cloud setting offers a practical trade-off: it reduces end-to-end latency by combining proximal preprocessing with cloud-based inference, which is particularly beneficial for time-sensitive monitoring pipelines.

Beyond raw latency and memory metrics, the cloud-native setup supports improved scalability under variable workloads. Automated scaling mechanisms help maintain throughput during peak influx periods, while modular deployment improves fault tolerance in production settings. Collectively, these results support the suitability of a cloud-ready, microservices-based infrastructure for operationalising epidemic forecasting pipelines in realistic public-health environments.

Beyond empirical latency measurements, we provide a lightweight complexity characterisation of the proposed pipeline. For each window of length  $W$  (with  $N = W$  nodes in the

corresponding visibility graph), visibility-graph construction has worst-case time complexity  $O(N^2)$  per window, while graph fusion and the extraction of structural descriptors scale with the number of retained edges. The prediction stage is computationally negligible: it uses a shallow regressor with fixed input dimensionality ( $d_{in} = 3$ ) and a small hidden layer ( $h = 10$ ), yielding  $O(d_{in} \cdot h)$  trainable parameters and  $O(d_{in} \cdot h)$  per-window inference cost. This cost decomposition supports the scalability claim, since per-window inference remains lightweight and can be parallelised efficiently in cloud environments. Notably, the regressor contains only 51 trainable parameters ( $3 \rightarrow 10 \rightarrow 1$ ), reinforcing the lightweight deployment footprint.

#### 4.7 | Summary of Findings

To evaluate the practical applicability and generalisability of the proposed model, we conduct a focused case study on regional epidemic forecasting. Unlike aggregate global trends, regional-level dynamics are subject to local policies, population density and healthcare responses, which introduce unique patterns and noise characteristics. By selecting a representative country or region with well-documented outbreak cycles, we compare the model's predicted trajectories with actual case counts over time. Special attention is paid to the model's ability to anticipate peaks, detect early surges and maintain consistency across shifting outbreak phases. This granular evaluation offers insight into

**TABLE 7** | Comparative summary of recent deep learning and graph-based models for epidemic forecasting.

| Reference         | Model type              | Data dependency                 | Handles stochasticity     | Dynamic graphs | Best reported RMSE | Interpretability |
|-------------------|-------------------------|---------------------------------|---------------------------|----------------|--------------------|------------------|
| Wang and Jin [26] | Graph transformer + SIR | Mobility data                   | Partial                   | Yes            | 0.0840             | Medium           |
| Cao et al. [24]   | GNN + mechanistic SIR   | Region-level cases              | No                        | No             | 0.0810             | High             |
| Mao et al. [23]   | Static + temporal GNN   | Influenza + COVID               | Yes                       | Yes            | 0.0780             | Medium           |
| Li & Ma [36]      | Transformer + GCN       | COVID global data               | Limited                   | Yes            | 0.0800             | Medium           |
| Ours (TGN-VG)     | VG + TGN + noise fusion | Confirmed case time series only | Yes (via noise modelling) | Yes            | <b>0.0709</b>      | High             |

the model's readiness for deployment in localised public health surveillance systems.

Moreover, the proposed model demonstrated consistent improvements in forecasting accuracy (MAE, RMSE, MAPE), exhibited strong resilience to noise and missing data and showed reliable performance even in short-term and long-term prediction settings. Compared to traditional and deep learning baselines, the temporal graph-based approach outperformed in nearly all configurations. Furthermore, the model's ability to integrate structural and stochastic information made it more adaptable and generalisable to diverse epidemic contexts. These insights provide both empirical evidence and theoretical justification for the adoption of this architecture in real-world epidemic forecasting pipelines.

Recent advances in epidemic forecasting have seen a shift from purely mechanistic models and standard machine learning methods to hybrid frameworks that integrate deep learning, GNNs and domain knowledge. As summarised in Table 7, multiple state-of-the-art models — such as M-Graphormer [26], MepoGNN [24] and BDSTGNN [23] — have contributed significantly by embedding epidemiological insights into neural architectures. However, most of these models either assume static or partially dynamic contact structures and few explicitly address the stochastic nature of real epidemic progression.

In contrast to prior methods that typically rely on deterministic representations or mobility-informed graphs, our proposed model introduces a unified framework that fuses three key innovations: (1) visibility graph construction to encode structural temporal dependencies, (2) stochastic noise modelling via fBm to simulate real-world randomness and (3) multi-layer temporal graph fusion to enrich dynamic representation. Unlike MepoGNN and STGNNs that require large-scale mobility or contact data, our approach remains robust with minimal assumptions and performs well even in data-sparse or noisy conditions.

From a qualitative perspective, compared to the mobility-aware deep learning models [29, 32, 38], our model eliminates dependency on external mobility data while still capturing time-evolving interaction patterns through visibility-based temporal

graphs. It also surpasses CNN-LSTM hybrids [35] in interpretability and generalisation due to its graph-structured encoding of epidemic dynamics. Furthermore, unlike MES-LSTM [37] or Bayesian-optimised models [40], which enhance performance via statistical tuning, our model inherently stabilises learning through topological and noise-aware graph design.

As presented in Table 3, the proposed TGN-VG model demonstrates competitive and often improved performance relative to the evaluated forecasting baselines under the reported experimental setting. While prior approaches such as M-Graphormer [26], BDSTGNN [23] and Transformer-GCN hybrids [36] integrate sophisticated graph structures or mechanistic modelling, they often rely on external mobility data or static assumptions. In contrast, our model uniquely combines visibility graph construction and stochastic noise fusion within a unified temporal learning architecture, enabling dynamic graph [47] adaptation without external dependencies. Notably, the TGN-VG model records the lowest RMSE (0.0709), outperforming competitors while preserving high interpretability — a rare combination in complex deep learning models. This comparative analysis highlights the model's readiness for real-world deployment, especially in uncertain and data-constrained environments.

Quantitatively, while some prior models such as M-Graphormer [26] and Li and Ma's transformer-GCN [36] achieved competitive RMSE values ( $\sim 0.08$ – $0.09$ ) on regional COVID-19 datasets, our approach achieves a lower RMSE ( $\sim 0.0709$ ) and MAE ( $\sim 0.0558$ ), along with stronger robustness under noise. Moreover, ablation studies reveal that even partial removal of the graph-based modules causes significant degradation, confirming their essential contribution — an insight not thoroughly explored in earlier literature.

In summary, while existing works have made meaningful progress in epidemic modelling through the use of GNNs, hybrid deep learning and domain-specific integration, the proposed method distinguishes itself through its fine-grained temporal graph modelling, stochastic resilience and independence from auxiliary data streams such as mobility or contact tracing records.

Although the experimental evaluation focuses on global COVID-19 incidence data, the proposed framework is not disease-

specific and operates on generic epidemic time-series signals. The visibility-graph construction and stochastic augmentation mechanisms depend only on temporal ordering and relative magnitudes, making them applicable to regional outbreak data and other infectious diseases with limited surveillance quality. In regional or non-COVID settings, the model can be adapted via fine-tuning on short historical segments or transfer-learning strategies, where graph-based representations learned from global data serve as an initialisation. A comprehensive evaluation across multiple diseases and geographic resolutions is left for future work.

The proposed framework is intended as a decision-support tool for public-health monitoring and operates exclusively on aggregated, population-level epidemic time-series data. It does not ingest individual-level records or personally identifiable information, thereby mitigating data privacy risks commonly associated with cloud-based analytics. From an interpretability perspective, the model is designed to be structurally transparent: predictions are derived from a compact set of graph-theoretic descriptors (average degree, clustering coefficient and graph entropy) extracted from visibility graphs, each of which has a clear and intuitive relationship to epidemic dynamics. This explicit feature design enables practitioners to qualitatively assess how changes in temporal structure and variability influence forecast outcomes, without reliance on opaque latent representations. While post-hoc explainability techniques such as attention visualisation or SHAP analysis are valuable for high-dimensional deep architectures, the low-dimensional and semantically meaningful feature space used here allows direct attribution of predictive behaviour without additional explanation layers. Finally, we emphasise that forecasts generated by the system should be interpreted as supportive signals rather than prescriptive outputs and should be considered alongside epidemiological expertise and contextual information when informing public-health decisions.

## 5 | Conclusion

In this study, we developed and evaluated a hybrid epidemic forecasting framework that integrates graph theory, temporal modelling, stochastic simulation and cloud-native deployment. By fusing visibility graphs derived from both empirical data and synthetic noise into a multilayer temporal graph, our approach effectively captures both the structured progression of epidemics and the inherent randomness and volatility of real-world conditions. The inclusion of biomedical semantics further enriches node-level features, enhancing interpretability and clinical relevance. Experimental results across diverse scenarios — including clean, noisy and incomplete datasets — demonstrate that the proposed model consistently outperforms classical statistical methods and state-of-the-art deep learning baselines. Notably, the model maintains predictive stability over long-term horizons and under data perturbations, underscoring its robustness and adaptability. Ablation studies confirm the critical contributions of each architectural component, particularly the visibility graph and noise fusion modules. Crucially, we implemented a cloud-native deployment architecture that significantly improves inference efficiency and resource utilisation. Cloud-based inference reduced execution time by over 60% and memory consumption by 35% compared to local CPU

deployment, enabling scalable, low-latency epidemic monitoring suitable for real-world public health applications. Overall, this work presents a principled, scalable and practical forecasting strategy that jointly addresses temporal dynamics, stochasticity, clinical semantics and operational feasibility. Future work will explore real-time multi-region deployment, integration of additional data modalities such as mobility and vaccination data and further optimisation of edge-cloud hybrid architectures.

## Author Contributions

Arman Kavooosi Ghafi (A.K.G.), Ali Pirkhedri (A.P.), Samira Akhbarifar (S.A.) and Mohammad Hossein Shafiabadi (M.H.S.) all contributed to the development of this study. A.K.G. and A.P. conceptualised the research idea and designed the overall methodology. A.K.G. performed the data preprocessing and initial experimental setup. A.P. implemented the core hybrid deep learning model and conducted the main experiments and performance evaluations. S.A. integrated biomedical imaging features into the framework and contributed to the validation and interpretation of results. M.H.S. supported the theoretical development in GNNs and stochastic modelling and contributed to manuscript revision. A.P. and A.K.G. wrote the main manuscript text. All authors reviewed and approved the final version of the manuscript.

## Funding

The authors have nothing to report.

## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

All datasets used in this study are publicly available and have been sourced from open-access repositories that are not under copyright restrictions. The primary dataset employed is the COVID-19 global confirmed case time series provided by the Johns Hopkins University Centre for Systems Science and Engineering (CSSE), accessible at: <https://github.com/CSSEGISandData/COVID-19>. All processed data and code necessary for replication are available upon request.

## References

1. S. Mahmud, H. Shen, Y. N. Z. Foutz, and J. Anton, "A Human Mobility Data Driven Hybrid GNN+RNN Based Model for Epidemic Prediction," in *2021 IEEE International Conference on Big Data (BigData)* (IEEE, 2021), 857–866, <https://doi.org/10.1109/BIGDATA52589.2021.9671474>.
2. M. S. Gawande, N. Zade, P. Kumar, S. Gundewar, I. N. Weeraratna, and P. Verma, "The Role of Artificial Intelligence in Pandemic Responses: From Epidemiological Modeling to Vaccine Development," *Molecular Biomedicine* 6, no. 1 (2025): 1, <https://doi.org/10.1186/s43556-024-00238-3>.
3. Y. Xiong, H. Wang, G. Liu, Y. Li, and T. Jiang, "Graph Neural Ordinary Differential Equations for Epidemic Forecasting," *CCF Transactions on Pervasive Computing and Interaction* 6 (2024): 281–295, <https://doi.org/10.1007/s42486-024-00161-0>.
4. R. Pastor-Satorras, C. Castellano, P. Van Mieghem, and A. Vespignani, "Epidemic Processes in Complex Networks," *Reviews of Modern Physics* 87, no. 3 (2015): 925–979, <https://doi.org/10.1103/RevModPhys.87.925>.
5. S. Shen, X. Hao, Y. Shen, et al., "Deep Reinforcement Learning-Based Botnet Propagation Control in the Social Internet of Things," *IEEE Internet of Things Journal* 12, no. 14 (2025): 27481–27495, <https://doi.org/10.1109/IIOT.2025.3562583>.
6. X. Zhu, Y. Zhang, H. Ying, H. Chi, G. Sun, and L. Zeng, "Modeling Epidemic Dynamics Using Graph Attention Based Spatial Temporal

- Networks,” *PLoS One* 19 (2024): e0307159, <https://doi.org/10.1371/journal.pone.0307159>.
7. J. Luo, X. Wang, X. Fan, et al., “A Novel Graph Neural Network Based Approach for Influenza-Like Illness Nowcasting: Exploring the Interplay of Temporal, Geographical, and Functional Spatial Features,” *BMC Public Health [Electronic Resource]* 25, no. 1 (2025): 408, <https://doi.org/10.1186/s12889-025-21618-6>.
  8. K. Chen, Z. Zhou, Y. Liu, et al., “TIGC-Net: Transformer-Improved Graph Convolution Network for Spatio-Temporal Prediction,” *Biomedical Signal Processing and Control* 100 (2025): 107024, <https://doi.org/10.1016/j.bspc.2024.107024>.
  9. M. Y. Arafat, S. Pan, and E. Bak, “An Adaptive Reinforcement Learning-Based Mobility-Aware Routing for Heterogeneous Wireless Body Area Networks,” *IEEE Sensors Journal* 24, no. 19 (2024): 31201–31214, <https://doi.org/10.1109/JSEN.2024.3440412>.
  10. J. Hoffmann and C. Caramanis, “Learning Graphs From Noisy Epidemic Cascades,” *Proceedings of the ACM on Measurement and Analysis of Computing Systems* 3 (2019): 1–34, <https://doi.org/10.1145/3309697.3331511>.
  11. A. Simone, A. Cesaro, and G. Esposito, “Analysis of the SARS-COVID-19 Trend: From Time Series to Visibility Graphs,” in *Proceedings of the 18th International Conference on Environmental Science and Technology (CEST)* (Global NEST, 2023), 1–4, <https://doi.org/10.30955/gnc2023.00202>.
  12. D. Tamir, N. Rishe, M. Last, and A. Kandel, “Soft Computing Based Epidemic Crisis Prediction,” in *Intelligent Methods for Cyber Warfare* (Springer, 2014), 43–67, [https://doi.org/10.1007/978-3-319-08624-8\\_2](https://doi.org/10.1007/978-3-319-08624-8_2).
  13. J. Mints, M. Rachdi, and J. Demongeot, “Stochastic Approach in Modelling Epidemic Spread,” in *2011 IEEE Workshops of International Conference on Advanced Information Networking and Applications* (IEEE, 2011), 478–482, <https://doi.org/10.1109/WAINA.2011.153>.
  14. S. Shubha, S. Mahmud, H. Shen, G. Fox, and M. Marathe, “Accurate and Efficient Distributed COVID-19 Spread Prediction Based on a Large-Scale Time-Varying People Mobility Graph,” in *2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS)* (2023), 58–68, <https://doi.org/10.1109/IPDPS54959.2023.00016>.
  15. D. Bucur and P. Holme, “Beyond Ranking Nodes: Predicting Epidemic Outbreak Sizes by Network Centralities,” *PLoS Computational Biology* 16 (2019): e1008052, <https://doi.org/10.1371/journal.pcbi.1008052>.
  16. P. Netrapalli and S. Sanghavi, “Learning the Graph of Epidemic Cascades,” in *Proceedings of the ACM SIGMETRICS/Performance Joint International Conference (SIGMETRICS '12)* (ACM, 2012), 211–222, <https://doi.org/10.1145/2254756.2254783>.
  17. H. Andersson and T. Britton, “Epidemics and Graphs,” in *Stochastic Epidemic Models and Their Statistical Analysis* (Springer, 2000), 63–72, [https://doi.org/10.1007/978-1-4612-1158-7\\_7](https://doi.org/10.1007/978-1-4612-1158-7_7).
  18. J. Hoffmann and C. Caramanis, “Learning Graphs From Noisy Epidemic Cascades,” in *Abstracts of the 2019 ACM SIGMETRICS/Performance Joint International Conference (SIGMETRICS/PERFORMANCE '19)* (ACM, 2019), 1–34, <https://doi.org/10.1145/3309697>.
  19. F. Mirsaeedi, M. Sheikhalishahi, M. Mohammadi, A. Pirayesh, and D. Ivanov, “Compartmental Models in Epidemiology: Bridging the Gap With Operations Research for Enhanced Epidemic Control,” *Annals of Operations Research* 357, no. 2 (2025): 1021–1078, <https://doi.org/10.1007/s10479-025-06893-1>.
  20. A. L. Costa, M. A. Pires, R. L. Resque, and S. S. M. S. Almeida, “Mathematical Modeling of Infectious Diseases: Key Concepts and Applications,” *Journal of Infectious Diseases and Epidemiology* 7 (2021): 209, <https://doi.org/10.23937/2474-3658/1510209>.
  21. S. K. Schnyder, J. J. Molina, J. C. Miller, R. Yamamoto, T. J. Kobayashi, and M. S. Turner, “Self-Organized Social Distancing During Epidemics When the Force of Infection Depends on Susceptible and Infectious Behavior,” *Mathematical Biosciences and Engineering* 23, no. 1 (2026): 148–171, <https://doi.org/10.3934/mbe.2026007>.
  22. M. C. González, C. A. Hidalgo, and A.-L. Barabási, “Understanding Individual Human Mobility Patterns,” *Nature* 453, no. 7196 (2008): 779–782, <https://doi.org/10.1038/nature06958>.
  23. J. Mao, Y. Han, G. Tanaka, and B. Wang, “Backbone-Based Dynamic Spatio-Temporal Graph Neural Network for Epidemic Forecasting,” *Knowledge-Based Systems* 296 (2024): 111952, <https://doi.org/10.1016/j.knsys.2024.111952>.
  24. Q. Cao, R. Jiang, C. Yang, Z. Fan, X. Song, and R. Shibasaki, “MepoGNN: Metapopulation Epidemic Forecasting With Graph Neural Networks,” *Machine Learning and Knowledge Discovery in Databases* 13718 (2023): 453–468, <https://doi.org/10.48550/arXiv.2306.14857>.
  25. M. Qiu, Z. Tan, and B.-K. Bao, “MSGNN: Multi-Scale Spatio-Temporal Graph Neural Network for Epidemic Forecasting,” *Data Mining and Knowledge Discovery* 38, no. 4 (2024): 2348–2376, <https://doi.org/10.1007/s10618-024-01035-w>.
  26. X. Wang and Z. Jin, “Multi-Region Infectious Disease Prediction Modeling Based on Spatio-Temporal Graph Neural Network and the Dynamic Model,” *PLOS Computational Biology* 21 (2025): e1012800, <https://doi.org/10.1371/journal.pcbi.1012800>.
  27. A. K. Ghafi, A. Pirkhedri, S. Akhbarifar, and M. H. Shafiabadi, “A Model for Predicting the Spread Patterns of Human and Computational Epidemics on Complex Temporal Networks,” *Applied Network Science* 10, no. 1 (2025): 1–31, <https://doi.org/10.1007/s41109-025-00752-5>.
  28. J. A. Miller, N. H. Barna, S. Rana, I. B. Arpinar, and N. Liu, “Knowledge Enhanced Deep Learning: Application to Pandemic Prediction,” in *2023 IEEE International Conference on Collaboration and Internet Computing (CIC)* (IEEE, 2023), 42–51, <https://doi.org/10.1109/CIC58953.2023.00016>.
  29. N. Neshat, S. Moayedfar, K. Rezaee, and N. A. Biuki, “Sustainable Planning of Developing Tourism Destinations after COVID-19 Outbreak: A Deep Learning Approach,” *Journal of Policy Research in Tourism, Leisure and Events* 16, no. 1 (2024): 1–21, <https://doi.org/10.1080/19407963.2021.1970578>.
  30. K. Rezaee, H. G. Zadeh, C. Chakraborty, M. R. Khosravi, and G. Jeon, “Smart Visual Sensing for Overcrowding in COVID-19 Infected Cities Using Modified Deep Transfer Learning,” *IEEE Transactions on Industrial Informatics* 19, no. 1 (2023): 813–820, <https://doi.org/10.1109/TII.2022.3174160>.
  31. Z. Shirmohammadi, K. Rezaee, M. Gheisari, et al., “Analysis of COVID-19 Prevalence in Belarus, Thailand, and Lithuania Utilizing the SIR Model during the COVID Era,” *Exploration of Medicine* 6 (2025): 1001331, <https://doi.org/10.37349/emed.2025.1001331>.
  32. K. Rezaee, A. Badii, and S. Meshgini, “A Hybrid Deep Transfer Learning Based Approach for COVID-19 Classification in Chest X-Ray Images,” in *2020 27th National and 5th International Iranian Conference on Biomedical Engineering (ICBME)* (IEEE, 2020), 234–241, <https://doi.org/10.1109/ICBME51989.2020.9319426>.
  33. A. Dairi, F. Harrou, A. Zeroual, M. M. Hittawe, and Y. Sun, “Comparative Study of Machine Learning Methods for COVID-19 Transmission Forecasting,” *Journal of Biomedical Informatics* 118 (2021): 103791, <https://doi.org/10.1016/j.jbi.2021.103791>.
  34. T. Zhan, Y. He, H. Li, and F. Xiao, “DVS: Deep Visibility Series and Its Application in Construction Cost Index Forecasting,” preprint, arXiv, November 7, 2021, <https://doi.org/10.48550/arXiv.2111.04071>.
  35. G. Yenurkar and S. Mal, “Future Forecasting Prediction of COVID-19 Using Hybrid Deep Learning Algorithm,” *Multimedia Tools and Applications* 82 (2022): 22497–22523, <https://doi.org/10.1007/s11042-022-14219-7>.
  36. Y. Li and K. Ma, “A Hybrid Model Based on Improved Transformer and Graph Convolutional Network for COVID-19 Forecasting,” *International Journal of Environmental Research and Public Health* 19, no. 19 (2022): 12528, <https://doi.org/10.3390/ijerph191912528>.
  37. T. Mathonsi and T. L. van Zyl, “A Statistics and Deep Learning Hybrid Method for Multivariate Time Series Forecasting and Mortal-

ity Modeling,” *Forecast* 4, no. 1 (2021): 1–25, <https://doi.org/10.3390/forecast4010012>.

38. F. Rahmadani and H. Lee, “Hybrid Deep Learning-Based Epidemic Prediction Framework of COVID-19: South Korea Case,” *Applied Sciences* 10, no. 23 (2020): 8539, <https://doi.org/10.3390/app10238539>.

39. M. R. Vargas, C. E. M. dos Anjos, G. L. G. Bichara, and A. Evsukoff, “Deep Learning for Stock Market Prediction Using Technical Indicators and Financial News Articles,” in *2018 International Joint Conference on Neural Networks (IJCNN)* (IEEE, 2018), 1–8, <https://doi.org/10.1109/IJCNN.2018.8489208>.

40. H. Abbasimehr and R. Paki, “Prediction of COVID-19 Confirmed Cases Combining Deep Learning Methods and Bayesian Optimization,” *Chaos, Solitons & Fractals* 142 (2020): 110511, <https://doi.org/10.1016/j.chaos.2020.110511>.

41. L. Wang, A. Adiga, J. Chen, A. Sadilek, S. Venkatramanan, and M. Marathe, “CausalGNN: Causal-Based Graph Neural Networks for Spatio-Temporal Epidemic Forecasting,” *Proceedings of the AAAI Conference on Artificial Intelligence* 36, no. 11 (2022): 12191–12199, <https://doi.org/10.1609/aaai.v36i11.21479>.

42. E. A. Radhi, M. A. Mohammed, F. Eid, and I. de la Torre Diez, “Efficient Blood Cell Classification Using a Hybrid Variational Quantum Circuit–EfficientNet Model on BloodMNIST,” *Journal of Artificial Intelligence in Medical Applications* 1, no. 1 (2025): 1–14, <https://doi.org/10.65511/jaima.v1i1.734>.

43. V. Capone, A. Casolaro, and F. Camastra, “Spatio-Temporal Prediction Using Graph Neural Networks: A Survey,” *Neurocomputing* 643 (2025): 130400, <https://doi.org/10.1016/j.neucom.2025.130400>.

44. G. Jin, Y. Liang, Y. Fang, et al., “Spatio-Temporal Graph Neural Networks for Predictive Learning in Urban Computing: A Survey,” *IEEE Transactions on Knowledge and Data Engineering* 36, no. 10 (2024): 5388–5408, <https://doi.org/10.1109/TKDE.2023.3333824>.

45. E. Dong, H. Du, and L. Gardner, “An Interactive Web-Based Dashboard to Track COVID-19 in Real Time,” *The Lancet Infectious Diseases* 20, no. 5 (2020): 533–534, [https://doi.org/10.1016/S1473-3099\(20\)30120-1](https://doi.org/10.1016/S1473-3099(20)30120-1).

46. Johns Hopkins University Center for Systems Science and Engineering, “COVID-19 Data Repository by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University,” Johns Hopkins University Center for Systems Science and Engineering, published March 10, 2023, <https://github.com/CSSEGISandData/COVID-19>.

47. Y. Zheng, L. Yi, and Z. Wei, “A Survey of Dynamic Graph Neural Networks,” *Frontiers of Computer Science* 19 (2025): 196323, <https://doi.org/10.1007/s11704-024-3853-2>.